



Service Catalog

Generated on: 2026-07-22 17:05:49 GMT+0000

SAP Advanced Therapy Orchestration | SHIP

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Original content: https://help.sap.com/docs/SAP_CGT_ORCH/1711ae9f34364a0880aaa6e52a9f6991?locale=en-US&state=PRODUCTION&version=SHIP

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Service Catalog

Capabilities of SAP Advanced Therapy Orchestration

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM001	Role template library	SAP Advanced Therapy Orchestration ships with 80 predefined role templates covering all applications functions. These templates serve as the foundation for creating tenant-specific roles. Customers create roles from these templates, optionally adding attribute values for data filtering, and bundle them into role collections assigned to users or user groups. In cell and gene therapy operations, where GxP-regulated organizations must document and justify every access decision, predefined templates significantly reduce the effort of designing a compliant authorization model from scratch and reduce the risk of inadvertently granting excessive access to patient-specific treatment data.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM002	4-level role separation	Every application function is split into four role levels (tile/launchpad access), Read (data viewing), Write (modification), and Action (specific operations like cancel, or book courier). This granular separation enables GxP-compliant segregation of duties, ensuring that who can view data cannot necessarily modify it, and who can modify it cannot necessarily approve it.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM003	Attribute-based data filtering	Roles can be configured with static attribute values to restrict data visibility. Seven filtering attributes are available: therapyId, countryId_code, plant, status, labelType, treatmentCenterId, and approvalType. This enables multi-therapy, multi-country, and multi-plant deployments where users only see data within their organizational scope. For cell and gene therapy companies managing multiple approved or investigational therapies across global manufacturing networks, attribute-based filtering ensures that site-level operators only see treatment orders and shipments relevant to their therapy, region, and plant – preventing accidental cross-contamination of operational context and supporting privacy requirements across jurisdictions.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM004	OR-logic aggregation	When a user is assigned multiple roles with different attribute values, the system aggregates access using OR logic. For example, a user with one role for Therapy A in Country X and another role for Therapy B in Country Y can see data for both combinations. This makes it straightforward to grant access across organizational boundaries without creating compound roles.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM005	Domain-specific config roles	Configuration administration is split into 10 granular roles across 5 domains (Sales, Material, Therapy, Org, and General Data), each with a Read and Write variant. These roles

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					the previous 3 monolithic configuration admin role release 2409, enabling tighter control over who can access which configuration areas. This was a breaking change that required customers to reassign roles during upgrade.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM006	Business role bundles	Three pre-packaged role collections align with the manufacturing personas: Biospecimen Manager, Finished Product Manager, and Process Order Manager. Each bundle includes tile access to the relevant set of manufacturing apps, simplifying role assignment for common user profiles without requiring customers to assemble individual role templates. These bundles map the distinct responsibilities in the CGT manufacturing workflow – biospecimen managers handle the collection to receipt phase, process order managers oversee manufacturing steps like transduction and expansion, and finished product managers manage the final product through to patient delivery – ensuring each person has precisely the access needed for their segment of the treatment lifecycle.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM007	Per-entity cancellation roles	Each shipment type (BS, BKS, IS, FS-IL, FS-FL) and treatment orders have dedicated cancellation request/approve/reject role pairs. This ensures that the authority to request a cancellation is separated from the authority to approve it, supporting GxP-compliant workflows. Cancellation roles for BKS, FS-IL, and IS were introduced in release 2502.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM008	Per-entity approval roles	Approval roles carry therapyId, countryId_code, and approvalType attributes, enabling fine-grained control over who can approve which entity types in which organizational contexts. This supports scenarios where different approvers are responsible for different therapies or regions, ensuring that approval authority matches organizational responsibility.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM009	PHI access isolation	Access to protected health information is controlled by a single dedicated role: CGT_PHI_Read_Srv. No other role in the system grants PHI access, enforcing the principle of least privilege. This role enables viewing of PHI-related documents across all transactional and manufacturing apps as well as the Monitor COI app, and must be explicitly assigned to each user who needs PHI visibility. This architectural decision reflects the heightened sensitivity of patient data in personalized therapies – unlike conventional drug manufacturing, CGT treatment courses are linked to individual patients, making PHI protection a core design concern rather than an ancillary compliance feature.
Application Platform & Infrastructure	Administration & Security	Authorization & Role Management	ARM010	Extension developer role	The ExtensionDeveloper role grants the ability to develop and activate custom extensions to a subscriber account. This is a technical role intended for implementation by SAP rather than business users. It is separate from all other roles.

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					templates to prevent accidental granting of deploy privileges to operational users.
Application Platform & Infrastructure	Administration & Security	E-Signature Management	ESM001	GxP-relevant step signing	SAP Advanced Therapy Orchestration requires electronic signatures for critical GxP-relevant operations including Change of Expiration Date, Manufacturing Termination, COI Data Update, and any change-and-save action shipments. When triggered, the user must re-authenticate with their e-signature credentials before the system accepts the change. This creates an auditable record of who authorized each GxP-critical action. In person-to-person medicine manufacturing, where every batch is unique to a patient, the consequences of an unauthorized expiration date change or erroneous COI update are direct patient safety risks – e-signatures ensure that each critical decision is deliberately and traceably authorized by a qualified person.
Application Platform & Infrastructure	Administration & Security	E-Signature Management	ESM002	OIDC-based authentication	E-signature uses a dedicated OpenID Connect (OIDC) for user re-authentication, separate from the main login. This requires a separate non-proxied SAP Identity Authentication Service (IAS) application specifically for signature users. Users must have verified accounts maintained passwords in SAP IAS, and the e-signature password is intentionally separate from SSO credentials to ensure true identity verification. The deliberate separation prevents a scenario where a user who left their workstation unlocked could have GxP-regulated actions performed under their identity – in CGT operations where a single misattributed action could affect patient traceability. An e-signature must represent a conscious, attributable decision.
Application Platform & Infrastructure	Administration & Security	E-Signature Management	ESM003	Third-party IdP support	SAP Advanced Therapy Orchestration supports a separate identity provider configuration where OIDC for e-signatures runs in SAP IAS while SAML for regular authentication runs in a third-party IdP such as Azure PingID, or Okta. This enables organizations to use their corporate IdP for day-to-day login while maintaining SAP IAS as the e-signature authority. Both federated scenarios (SAP IAS only, or SAP IAS + third-party) are fully supported.
Application Platform & Infrastructure	Administration & Security	E-Signature Management	ESM004	Extension field e-signature	When users update custom extension fields on biospecimen or finished product shipments, the system prompts for e-signature validation just as it would for standard field changes. This ensures that customer-specific data extensions are held to the same GxP compliance standards as SAP-delivered fields. This capability was introduced in release 2404.
Application Platform & Infrastructure	Administration & Security	E-Signature Management	ESM005	Configurable e-signature endpoint	The e-signature authentication endpoint defaults to IDP_OIDC_URL HTTP destination but can be overridden via the ESIGNATURE_AUTH_URL rule provider. This allows customers to point e-signature authentication to a different endpoint.

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					IAS instances or configurations based on their organizational setup, providing flexibility in multi-tenant multi-region deployments.
Application Platform & Infrastructure	Administration & Security	E-Signature Management	ESM006	E-signature audit trail	Every e-signature attempt is recorded in the SAP A with three possible outcomes: Try (initiated), Success (completed), or Failed (rejected). These events are the Security Events audit category and are retained 90 days in the audit log. For long-term GxP compliance, customers must archive these records externally using the Audit Log Retrieval API. This comprehensive logging including failed attempts that may indicate unauthorized access – provides the evidence trail that regulatory inspectors expect during GMP audits of CGT manufacturing processes.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL001	Audit logging (3 types)	SAP Advanced Therapy Orchestration records three categories of audit events: Personal Data Changes (tracked via Personal Data Manager), Configuration Changes (creation and activation/deactivation of configuration records), and Security Events (e-signature attempts and sensitive data access such as document viewing). All three categories write to the SAP Audit Log service and cannot be disabled by users. In CGT manufacturing, where every treatment order is link specific patient, these audit categories collectively form the three pillars of GxP compliance – protecting patient data, controlling system configuration, and tracking accessed or authorized sensitive information.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL002	90-day audit log retention	The SAP Audit Log service retains events for 90 days with no option to extend this period within the service. In GxP-regulated environments that require longer retention, customers must implement external archiving using the SAP Audit Log Retrieval API. SAP security recommendation CGTO-0002 explicitly advises downloading and archiving audit logs regularly to meet compliance requirements.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL003	Audit Log Viewer	The SAP Audit Log Viewer provides a UI for browsing and searching audit events. Access requires a subscription to the SAP Audit Log Viewer service on BTP and assignment of the auditlog-viewer!t*/Auditlog Auditor role collection. The viewer supports filtering by event type, time range, and user, enabling compliance officers to investigate specific activities.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL004	Change logs (indefinite)	Business-level change logs track all modifications across all entities and persist indefinitely – they are the primary compliance audit trail, separate from the 90-day audit logs. Every change captures the attribute modified, the action taken, old and new values, who made the change, when, and from which source system. Change logs cannot be deleted, modified, or stopped by any user. In cell and gene therapy, where a single treatment order can span

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					months and involve dozens of handoffs across collection sites, manufacturing plants, and distribution centers. Indefinite retention of change logs ensures that the complete decision history is available for regulatory inspection, deviation investigation, or patient safety at any point in the future.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL005	Change log hierarchy	Starting with release 2502, change logs display a Parent Entity column showing the top-level entity name and business key. Release 2506 enhanced this further by displaying business keys for both the Parent Entity and Current Entity columns, enabling users to trace changes through the full entity hierarchy (e.g., Shipment 14! Shipment Material 10 -> Material Subunit 20). This hierarchical traceability is essential in CGT because treatment orders contain deeply nested entity structures. A single change to a material subunit's batch disposition can have upstream implications for the shipment, the processing activity, and ultimately the patient's treatment timeline.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL006	Change log attributes	Each change log entry captures a comprehensive set of attributes: the specific attribute changed, the action (Create or Update), the Change Reason (type and description), the old value, the new value, the user who made the change (Changed By), the timestamp (Changed On), and the source system identifier (Changed By System). This level of detail supports full traceability for GxP audits.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL007	View Change Logs app	The central View Change Logs app provides a unified view of all changes across the system. The default display period is 10 days (configurable), with a maximum of 300,000 records per query. Results are sorted chronologically and can be exported to Excel. Users can also view change logs per record within each individual entity for context-specific auditing.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL008	Root Entity column	Introduced in release 2502, the Root Entity column displays the name and business ID of the top-level entity for every change log entry. This makes it possible to quickly identify which shipment, order, or processing activity a change belongs to, even when the change was made to a deeply nested sub-entity. Data enrichment for records created before 2502 was applied retroactively.
Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL009	Extension field change logs	When custom extension fields are created or updated, either manually through the UI or via inbound event messages, the system automatically creates change log entries. This applies to biospecimen shipments and collection shipments, ensuring that customer-specific data extensions have the same audit trail as standard fields. Introduced in release 2404.

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Application Platform & Infrastructure	Administration & Security	Audit & Change Log Management	ACL010	Non-stoppable logging	Both change logs and audit logs are designed to be tamper-resistant: they cannot be stopped, paused, modified by any user, including system administrators. This is a deliberate GxP design choice ensuring the compliance trail is always complete and cannot be circumvented. The only user control is the display of the View Change Logs app.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM001	Monitor Business Events app	The Monitor Business Events app provides a central view of all inbound and outbound business events processed by SAP Advanced Therapy Orchestration. Each event is tracked by its status, payload, timestamps, and processing details. This is the primary tool for system administrators to monitor, diagnose integration issues, verify event processing, and understand the flow of data between SAP Advanced Therapy Orchestration and external systems. Because treatment orchestration depends on dozens of event-driven integrations with ERP, ordering platforms, and external manufacturing partners, rapid visibility into failures is critical – a stalled Purchase Order creation could delay biospecimen pickup and jeopardize the viability window for a patient's cells.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM002	Event retrigger	Failed events can be retriggered directly from the Monitor Business Events app. Retriggering is supported for events with error code ES004 (technical or functional failure), ES005 (backend service down), as well as events that have been stuck in Processing or Transformed status for more than 15 minutes. Non-retriggerable errors (ES001 unknown event type, ES002 missing mandatory field, ES003 parsing failure) require fixing the root cause and resubmission. In time-sensitive CGT workflows where viability windows may be as short as 24-72 hours, the ability to quickly retrigger failed events without resubmitting from the source system can be the difference between a successful treatment and a lost therapy opportunity.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM003	PPF event correlation	The Post Processing Framework (PPF) assigns a Correlation ID to each event it triggers, linking inbound events to the resulting outbound business events. This correlation is trackable across the Manage Orders, Shipments, and Manage Processing Activity apps, enabling end-to-end tracing of how a business action (e.g., sales order confirmation) resulted in a specific outbound event (e.g., Sales Order Create). This traceability is particularly valuable during deviation investigations, where quality teams need to reconstruct the exact sequence of system actions that led to a specific ERP document being created for a patient's treatment.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM004	Business logs	The Monitor Business Events app captures logs for external synchronous API calls including document creation and upload operations, label generation and validation requests, patient details retrieval, and courier booking requests. These logs provide visibility into the real-time

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					integrations that SAP Advanced Therapy Orchestra performs with external services, complementing the asynchronous event monitoring. Additional synchronous APIs like Get Slots & Book Slots are also available in the app
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM005	Event updates	Introduced in release 2409, event updates show descriptive messages with timestamps when notable conditions occur during event processing. This includes scenarios such as duplicate Event ID detection (when a previously used event ID is resubmitted) and event sequencing waits (where an event is held until a prerequisite event completes). The update payload is viewable when an event is sent with a previously used event ID.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM006	Outbound workflow visibility	Synchronous outbound API calls from SAP Advanced Therapy Orchestration to SAP Build Process Automation workflow instances are classified as OUTBOUND_SYNCHRONOUS in the Monitor Business Events app. Users can view both the payload sent to the workflow API endpoint and the received response. This feature was introduced in release 2506 to support the shipper create/update workflow trigger feature. Only workflow calls are covered; other synchronous outbound calls such as those to Business Rules are not included.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM007	Real User Monitoring	SAP Advanced Therapy Orchestration integrates with Cloud ALM for Real User Monitoring (RUM), which measures usage patterns and performance of the application. Currently this covers the Manage Order microservice. RUM uses OpenTelemetry auto-instrumentation and SAP Passport for request correlation, providing insights into response times, error rates, and user experience metrics.
Application Platform & Infrastructure	Administration & Security	Business Event Monitoring	BEM008	Integration Monitoring	SAP Cloud ALM Integration Monitoring provides end-to-end visibility across all microservices, tracking inbound and outbound events as well as synchronous API calls. It supports filtering by Status Group, Message Direct Event Type, enabling administrators to identify bottlenecks or failures in the integration landscape. This complements the per-event view in the Monitor Business Events app with a system-wide perspective.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM001	Full transport	A full transport exports all configuration entities and their records from one tenant and imports them into another. This is typically used for initial setup of a new tenant (pre-production or production) where the entire configuration baseline needs to be replicated. The transport is executed via SAP Cloud Transport Management and creates corresponding change log entries in the destination tenant. In GxP-validated environments, configuration consistency between development, pre-production, and production tenants is a regulatory requirement.

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					requirement – full transport ensures that the validated configuration baseline is replicated exactly, eliminating manual transcription errors that could lead to the misconfiguration.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM002	Custom transport	A custom transport allows administrators to select configuration entities and specific records within those entities for transport. This is useful for incremental changes where only certain configuration areas have been modified, avoiding the overhead of a full transport. Administrators choose which entities and records to include before creating the transport request.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM003	Transport templates	Introduced in release 2404, transport templates are reusable definitions of which configuration entities and records should be included in a transport. Templates can be created for specific therapy or country rollouts, activated or deactivated, and copied to create variations. This dramatically reduces the time required for multi-therapy or multi-country deployments – when a pharmaceutical company launches a new CGT therapy in additional markets, a pre-built transport template ensures the validated configuration is applied consistently to each country rollout without recreating the transport set from scratch.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM004	Manage Transports app	The Manage Transports app provides a UI for creating, viewing, and managing transport requests. Release 2605 enhanced the app with three new fields: ID (a unique system-generated identifier for tracking requests across systems), Transport Type (Full or Custom), and Status (Successful or Failed with error details). The Incoming Transports and Outgoing Transports tabs provide views into both sent and received transports.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM005	Transportable entities	Approximately 47 configuration entity types can be transported, covering the core configuration areas: Organizations, Distribution Channels, Divisions, Sales Areas, Plants, Order Types, Business Partner Roles, Address Types, Therapy Categories/Types, Material Types/Groups/Combinations, Purchasing Organizations, Indications, EAN Categories, Temperature Conditions, Codes, Transportation Groups, Packaging Product Types, Transportation Lane Rules, Reason Codes, Change Reasons, Status Profiles, User Statuses, Document Types, Organization Location Types, Location Rules, Post-Processing Events, Field Control Profiles, Milestone Types, Materials (BS/FP), Flow Section Types, and Groups. 12 entities added in 2605: entity status processing activity types, rules, destinations, customer types, COI generation rules, COI/COC event profile external events/attributes, COI certificate templates, and courier instructions.

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Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM006	Non-transportable entities	Certain configuration entities are intentionally excluded from transport because they are tenant-specific by design. These include: Number Ranges, COI Generation Rules, Cloud Services, Rules, Destinations, Entity Status Fields, Processing Activity Types, Custom Text Types, COI/Event Profiles, External Events/Attributes, COI Certification Templates, and all Master Data apps. These must be configured manually in each tenant. This deliberate exclusion prevents transporting tenant-specific settings such as COI generation rules or system destinations. Configuration must be uniquely configured per environment to avoid numbering collisions, incorrect service endpoints, and misrouted patient identity data.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM007	Transport of sub-entities	Automatic sub-entity transport with parent entities. Full Transport and Custom Transport automatically transport sub-entity data when transporting configuration entities, eliminating manual recreation requirements in target tenants. Supports multi-level sub-entity hierarchies. Validates dependent sub-entities exist in target before import – if validation fails, import request fails with detailed error messages. Change logs created for sub-entity updates. Affected entities: Configure Order Types (follow-on order types), Configure Change Reasons, Configure Reason Codes (reason codes), Configure Profiles (status mappings), Configure Post Process Events (events), Configure Profile Groups (shipment/document/processing activity profiles), Configure Field Control Profiles (attributes), Configure Courier Instructions (order types). Sub-entities automatically included based on parent selection; not visible in Select/View Record screens. Enhanced from manual recreation to automatic transport.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM008	Change log on import	When configuration data is imported into a destination tenant, the system automatically creates change log entries for all imported records. This ensures that the configuration audit trail is maintained even for transported data, documenting when the import occurred and which records were affected. This capability was introduced in release 2404.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM009	SAP Cloud Transport Management	SAP Cloud Transport Management is the underlying service that executes the actual transport of configuration data between tenants. It manages the transport layer (source -> destination mappings), handles the file transfer and provides status tracking. Administrators can view transport request details on the Content tab of the Transport Management dashboard.
Application Platform & Infrastructure	Administration & Security	Configuration Transport Management	TPM010	Download/Upload Entity APIs	Two REST APIs enable programmatic master data transport between tenants: POST /downloadEntity (exports master data as JSON) and POST /uploadEntity (imports master data as JSON). These APIs use session keys combining parent and child business keys for

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					identification. Deletions in the source are intention propagated to the target for data integrity. Release massively expanded coverage: Download Entity gai ~30 configuration data entities and 8 master data Upload Entity gained 7 master data entities.
Application Platform & Infrastructure	Administration & Security	SAP Build Work Zone	BWZ001	Unified Fiori Launchpad	SAP Build Work Zone enables customers to access Advanced Therapy Orchestration applications from unified Fiori Launchpad alongside their SAP S/4HANA apps, custom-built applications, and third-party extensions. This provides a single point of entry for who work across multiple SAP products, eliminating need to switch between separate launchpads. For C operations teams that frequently move between the order management in CGTO and ERP transactions S/4HANA (such as reviewing purchase orders, process orders, or goods movements), a unified launchpad context-switching overhead and accelerates response times for time-sensitive therapy workflows. Introduced in release 2409.
Application Platform & Infrastructure	Administration & Security	SAP Build Work Zone	BWZ002	SAP Task Center integration	When organizations implement workflows in SAP E Process Automation and a workflow task is assigned an approver, the task is also visible in SAP Task Center SAP Build Work Zone. This allows approvers to view process tasks from a central location alongside tasks from other SAP applications. The system administrator can configure SAP Advanced Therapy Orchestration, SAP E Process Automation, and SAP Task Center together in the Build Work Zone. This was introduced in release 25
Application Platform & Infrastructure	Administration & Security	SAP Build Work Zone	BWZ003	Cross-subaccount access	SAP Advanced Therapy Orchestration applications accessed from BTP subaccounts that are not directly subscribed to SAP Advanced Therapy Orchestration through the SAP Build Work Zone site manager. For cross-subaccount scenarios, it is recommended to “Open in a New Tab” option rather than the “In-place” option when configuring app access, to avoid navigation issues.
Application Platform & Infrastructure	Administration & Security	SAP Build Work Zone	BWZ004	Limitations	When SAP Advanced Therapy Orchestration apps are accessed through SAP Build Work Zone rather than the native CGTO launchpad, certain features are not available: Built-in Support (incident creation and expert chat), app Help system, and the Consent management functionality. These features require the native CGTO launchpad context and cannot be surfaced through the Build Work Zone integration layer.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI001	Purchase Order lifecycle	When a biospecimen shipment schedule is confirmed in SAP Advanced Therapy Orchestration triggers a Purchase Order create request to SAP S/4HANA via SAP Event Manager and SAP Integration Suite (CPI) with predefined field mappings. The full lifecycle is supported: creation, response, collection update, schedule update, local

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					change, and cancellation (logical deletion of PO item CGTO data (COI ID, shipment ID, collection info, sub details) is displayed and editable on the PO screen transactions ME21N/ME22N. The PO represents the procurement of a patient's biological starting material typically an apheresis collection – from the treatment center, making it the ERP anchor for the first physical in the CGT supply chain.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI002	Sales Order lifecycle	Schedule confirmation for finished product final leg biospecimen kit shipments triggers Sales Order creation in SAP S/4HANA. The lifecycle includes creation, update, and cancellation events, with response events from SAP S/4HANA back to SAP Advanced Therapy Orchestration. Batch allocation events are triggered when picking assigns a batch to the outbound delivery, and deallocation events fire when picked quantities are removed. Document ID, Study ID, and Protocol ID are included in the event (2506). The SO represents the delivery of the manufactured therapy product to the patient's treatment center – the final physical step before infusion – making batch allocation accuracy a direct patient safety concern.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI003	Stock Transport Order lifecycle	Schedule confirmation for intermediate product arrival at finished product transit leg shipments triggers STO creation in SAP S/4HANA. The full lifecycle covers creation, update, and cancel requests with response events. Event data (sending/receiving plants, create/update/cancel responses) is displayed on the ERP tab in the Manage Shipments apps. Business partner updates on IS orders and shipments are synced to SAP S/4HANA via STO update events. Introduced in release 2502.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI004	Manufacturing Order lifecycle	Processing activities map to either process orders or production orders (PP) in SAP S/4HANA, configured via order type via the Manufacturing Order Type field introduced in release 2506. The lifecycle covers creation, update, and cancellation with response events. The order type is determined via a BRF rule, and the order can be derived from the treatment order type. Default behavior creates a process order if the manufacturing order type is not configured. This flexibility accommodates the complex CGT manufacturing landscape where some organizations use process industry (PI) workflows for cell therapy expansion and others use discrete manufacturing (DM) for gene therapy vector production.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI005	Goods Issue and Reversal	Post Goods Issue events from SAP S/4HANA update the corresponding finished product shipment in SAP Advanced Therapy Orchestration. When a PGI is reversed, the reversal event removes handling units from the CGT for the corresponding subunit items. PGI is supported for sales orders and stock transport orders. The PGI/reversal events carry CGTO data (COI ID, shipment ID, subunit

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					information, handling unit IDs) through the complete document chain.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI006	Goods Receipt and Reversal	Post Goods Receipt events update biospecimen shelf life (from purchase orders) and processing activities (from process/production orders) in SAP Advanced Therapy Orchestration. PGR carries batch numbers, vendor numbers, and batch expiry dates (calculated from production date + shelf life in material master, added in 2506). The actual delivery date of the biospecimen shipment is updated with the material document date. PGR reversal retains previously existing batch and batch values in SAP Advanced Therapy Orchestration.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI007	ERP document data flow	CGTO data flows seamlessly through the complete S/4HANA document chain: Purchase Order -> Inbound Delivery -> Post Goods Receipt, and Sales Order -> Outbound Delivery -> Post Goods Issue. COI ID, shipment ID, subunit info, and handling units are auto-copied at each stage. Subunits are editable on the PO screen; handling units are manually maintained on delivery screens. Data is also copied between documents (SO->PO, PO->PO). Leading zeros are removed from all outbound payloads (2409). This end-to-end data propagation provides the foundation for maintaining Chain of Identity across the ERP layer – without it, warehouse and logistics personnel would need to manually re-enter patient-specific identifiers at each document stage, creating an unacceptable mix-up risk in autologous therapy supply chains.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI008	CGTO Information tab in SAP S/4HANA	SAP S/4HANA documents display CGTO data on a dedicated CGTO Information tab. This includes purchase orders, sales orders, process orders, production orders (CO01/CO02/CO03, added in 2506), stock transport orders, inbound/outbound deliveries, and material documents (MIGO). Fields include COI ID, shipment Donor ID, Study ID, Protocol ID, and subunit details. Production order and process order tabs show component materials and subunit data referenced by line item number.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI009	Production milestones	Manufacturing start, end, and termination events are captured in SAP S/4HANA (via transactions COR6/COR6N, CORZ, CORK, CORS) and sent to SAP Advanced Therapy Orchestration to update the corresponding processing activity. Consumption and production events during production order processing also flow from SAP S/4HANA to SAP Advanced Therapy Orchestration. Both process orders and production orders are supported (production order support added in 2506). Milestone events include actual processing start and end dates.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP	EDI010	Batch expiry date	During Post Goods Receipt for a process order or production order, the expiry date in the material document is populated with the Shelf Life Expiration or Best-Use date.

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		Document Integration			Date, calculated by adding the production date and shelf life from the material master. This expiry date passed to SAP Advanced Therapy Orchestration via PGR event. If the expiry date is received via the event CGTO-side logic for calculating expiry date based on Processing Date is not triggered. Accurate expiry propagation is critical for CGT products, which often have very short shelf lives – some non-cryopreserved cell therapy products must be administered within hours, making automated expiry calculation a patient safety imperative. Introduced in release 2506.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI011	CPI iFlow extensions	All existing CPI integration flows (iFlows) in the SAP Advanced Therapy Orchestration integration package have been enhanced with both pre-exit and post-exit exit points. This allows customers to modify messages before and after standard mapping and execution, and to make outbound calls to different receivers via custom routing logic. Customers must download or re-import iFlow from SAP Business Accelerator Hub to get the extension modifications to imported iFlows will be overwritten on update. Introduced in 2506.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI012	COI document lookup BAPI	The /CGTO/GET_COI_ASSOC_DOCS RFC function in SAP S/4HANA accepts one or multiple COI IDs and returns all associated SAP S/4HANA documents and their CGTO data (including subunit information). This native API enables customers to build COI-based document reports directly in SAP S/4HANA without querying SAP Advanced Therapy Orchestration. In operations, the ability to trace all ERP documents linked to a specific patient's COI ID is essential for regulatory inquiries, deviation investigations, and batch release reviews where quality teams need to reconstruct the complete document trail for a single patient's therapy. Introduced in release 2409.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI013	CGTO Event Monitor	Transaction /CGTO/EVENT_MONITOR in SAP S/4HANA displays all CGTO-related inbound and outbound events with their processing statuses. Administrators can acknowledge outbound events that need to be reprocessed. Access requires authorization object /CGTO/EVMO (activity 02 for replay, activity 03 for display). This provides visibility into the SAP S/4HANA event flow, complementing the Monitor Business Events app on the SAP Advanced Therapy Orchestration side. Introduced in release 2404.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI014	SAP S/4HANA Add-On for Life Sciences	Starting with SAP Advanced Therapy Orchestration 400, the CGTO functions formerly provided by the standalone SAP S/4HANA Add-On for SAP Cell and Gene Therapy Orchestration have been consolidated into SAP S/4HANA Add-On for Life Sciences (ISLS). The current version is ISLS 400/FPS02. After installation or upgrade, the business functions switch must be explicitly enabled.

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					off by default). The add-on has evolved through 5 v FPS02 (2404), FPS03 (2409), FPS04 (2502), ISLS 400/FPS01 (2506), ISLS 400/FPS02 (2511).
Application Platform & Infrastructure	Integration	SAP S/4HANA Warehouse Management Integration (EWM)	EWM001	HU-to-SU sequential mapping	Handling units (HUs) from SAP S/4HANA Embedded Extended Warehouse Management are mapped to subunits (SUs) in SAP Advanced Therapy Orchestration using sequential mapping: the 1st HU maps to the 1st SU, the 2nd HU to the 2nd SU, and so on. If the number of HUs does not match the number of SUs, the system logs an error and does not trigger the event. This mapping applies across purchase orders, process orders, sales orders, and stock transport orders, and works across multiple PGR documents. Accurate HU-to-SU mapping is fundamental to maintaining Chain of Identity in the warehouse – each subunit represents a patient-specific material portion, and a mapping error could result in the wrong patient's therapy product being picked, packed, and shipped.
Application Platform & Infrastructure	Integration	SAP S/4HANA Warehouse Management Integration (EWM)	EWM002	Partial quantities for serialized materials	Comprehensive support for partial goods movement of serialized materials across all shipment types, introduced in release 2511. Biospecimen shipments support partial/full consumption via process/production order events. Finished product shipments (both final leg and transit leg) support partial allocation/deallocation and processing activities. Intermediate product shipments support partial PGI/PGR reversal, disposition, and allocation/deallocation. Processing activity-to-process activity flows support partial consumption/non-consumption of HU/subunit. This capability addresses common CGT manufacturing reality where a single biospecimen collection may be split across multiple processing steps or where a manufacturing run yields partial quantities, requiring the warehouse to move and track individual handling units independently rather than in bulk.
Application Platform & Infrastructure	Integration	SAP S/4HANA Warehouse Management Integration (EWM)	EWM003	Batch status transitions	Six batch status transition scenarios are supported for quality inspections via EWM transfer postings: Blocked ↔ Unrestricted, Blocked ↔ Quality Inspection, and Quality Inspection ↔ Unrestricted. When a transfer posting is created in SAP S/4HANA EWM, disposition events are sent to SAP Advanced Therapy Orchestration to update the corresponding entity (biospecimen, finished product, intermediate product, or transit leg shipment). This integration enables real-time batch status synchronization between the warehouse and the orchestration platform. Introduced in release 2511.
Application Platform & Infrastructure	Integration	SAP S/4HANA Warehouse Management Integration (EWM)	EWM004	Event sequencing	When goods movement and stock posting occur simultaneously in EWM for the same batch, multiple outbound events may fire at once. The /CGTO/RESET_MATDOC_EVENTS report sequences events using group and sequence identifiers, ensuring

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					Advanced Therapy Orchestration processes them in the correct order. A background job /CGTO/EWM_OBD_EVENT schedules this automatic cleanup and /CGTO/MATDOC_CLEANUP handles log cleanup and prevents race conditions that could lead to incorrect status updates. Introduced in release 2511.
Application Platform & Infrastructure	Integration	SAP S/4HANA Warehouse Management Integration (EWM)	EWM005	COI ID validations during goods movements	Robust COI ID validations are performed during key movement steps in SAP S/4HANA to ensure the therapy is administered to the correct patient. Validation points include: biospecimen receipt at plant, transfer from storage to production, process/production order receipt, biospecimen material consumption, batch picking delivery, and batch shipment. The batch characteristic name for COI ID is configurable, and validation messages can be overridden from error to warning or info severity. Non-blocking when COI ID is not found in document. Introduced in release 2511.
Application Platform & Infrastructure	Integration	SAP S/4HANA Warehouse Management Integration (EWM)	EWM006	MIGO enhancements	The MIGO transaction in SAP S/4HANA has been enhanced with CGTO-specific capabilities. COI reference fields are enabled for Movement Type 343 (Disposition) and Movement Type 322 (Reversal), with both ever transmitted to SAP Advanced Therapy Orchestration for traceability. Additionally, material document display MIGO shows read-only CGTO data (COI ID, shipment subunit info) for goods receipts, goods issues, and reversals related to CGTO process orders, sales or purchase orders.
Application Platform & Infrastructure	Integration	SAP Transportation Management Integration	TMI001	CGTO data in freight objects	When a freight unit is created against a sales order or purchase order, or stock transport order line item in SAP S/4HANA Transportation Management, CGTO information (COI ID, shipment ID) is automatically copied from SAP S/4HANA document to the freight unit, and then propagated to the freight order and freight booking. This ensures that CGTO traceability data is available throughout the transportation management process. CGTO data in freight objects is not currently displayed in the TM user interface. Extending COI traceability in freight management ensures that even during transportation – where temperature excursions, delays, and misrouting pose direct risks to cell viability – the precise specific identity of the shipment is preserved across all logistics documents. Introduced in release 2502.
Application Platform & Infrastructure	Integration	SAP Transportation Management Integration	TMI002	Courier Booking Response from TM	When courier information is updated on a Freight Confirmation, Freight Booking in SAP S/4HANA Transportation Management, real-time courier data is sent to SAP Advanced Therapy Orchestration. The payload includes waybill information, Account Number, and Carrier ID. The trigger only fires if the Freight Order/Booking has a valid COI ID, preventing false positives from non-CGTO freight activities. The extension framework in the SAP S/4

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					add-on allows customers to customize the data model. Introduced in release 2502.
Application Platform & Infrastructure	Integration	SAP Transportation Management Integration	TMI003	Courier Booking Milestone Update	When proof of delivery (VLPOD) is executed in SAP S/4HANA Transportation Management, milestone information is sent from the SAP S/4HANA add-on to Advanced Therapy Orchestration. The courier interface has been enhanced to capture multiple milestones and their attributes. Currently only the VLPOD milestone is built into the standard delivery, but the framework is extensible via the SAP S/4HANA add-on to support additional milestone types. Delivery confirmation milestones are particularly important for CGT shipments because treatment centers need real-time visibility when patient-specific therapy products arrive, enabling them to schedule the infusion procedure within the product's viability window. Introduced in release 2502.
Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC001	Post Processing Framework (PPF)	The internal event engine of SAP Advanced Therapy Orchestration with ~70 business contexts and ~50 configurable events that trigger follow-on processes when business objects change. PPF events drive ERP document creation, workflow triggering, COI event recording, exception generation, and schedule confirmation. Configuration is managed via the Configure Post-Processing Events app, where business contexts are mapped to specific events. Critical insight: automatic flow validation, confirmation and automatic schedule confirmation are driven by PPF events that must be explicitly configured – they are not built-in defaults. PPF is the central nervous system of SAP Advanced Therapy Orchestration's orchestration layer, virtually every automated action in the treatment lifecycle, from triggering a Purchase Order when a biospecimen schedule is confirmed to recording a COI event when a goods receipt is posted, depends on correctly configured PPF events.
Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC002	SAP Event Mesh messaging	SAP Event Mesh provides the asynchronous messaging layer between SAP Advanced Therapy Orchestration and external systems using AMQP/HTTP protocols. Inbound events are received on a single topic (<code><namespace>/inbound/events/receiver</code>), while outbound events are routed to per-business-object topics (<code><namespace>/outbound/<business-object></code> ; e.g., Order, Process Order, Purchase Order, Stock Transfer Order, Text, and Miscellaneous). Inbound responses use a dedicated topic (<code><namespace>/ce/inbound/responses</code>) requiring a subscription. The event catalog spans ~100+ event types across 13 functional domains. This event-driven architecture is fundamental to CGT orchestration, as treatment order processing involves continuous data exchange between the ordering platform, ERP, manufacturing systems, and logistics providers – all of which must stay synchronized to maintain the integrity of each patient's individualized therapy workflow.

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Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC003	Cloud Event format	All events use the Cloud Event specification (version with standardized headers: CE-Specversion, CE-Source (namespace), CE-Type (topic for routing), CE-ID (unique identifier), and qos (AT_MOST_ONCE quality of service). Every inbound cloud event has a corresponding response event of the same event type. Events are published to SAP Event Mesh at the URL {event-mesh-url}/sap/ems/v1/events. This standardized format ensures interoperability with any Cloud Event-compatible messaging system.
Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC004	OpenAPI to Cloud Events migration	Cloud Events became the default event format in SAP Advanced Therapy Orchestration 2506. OpenAPI events are deprecated, with support continuing through release 2605 and decommissioning in release 2611. To migrate customers create a support case with SAP to toggle the ft_cloud_events feature flag, then prepend /ce to all topics. By default, all events are already processed in Cloud Event format, so migration primarily affects customers who built integrations on the OpenAPI format.
Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC005	Multiple Treatment Center Platform routing	BRFplus-based routing of outbound events to different Treatment Center Platform (TCP) instances, introduced in release 2409. The routing uses the eventTargetSystem parameter and is determined based on Therapy, Center, and Treatment Center, and Order Type. This allows switching between TCP instances during end-to-end orchestration (e.g., different instances for commercial vs. clinical). Supports 13 outbound event types but is not supported for SAP S/4HANA-related outbound events. This routing flexibility is critical for pharma companies that use different ordering platforms for commercial versus therapy programs, or that operate separate TCP instances per region to comply with data residency requirements.
Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC006	Event error handling	Five event error codes define the processing outcome: ES001 (unknown event type), ES002 (missing mandatory field), ES003 (parsing failed), ES004 (action failed technically/functional), and ES005 (backend service unavailable). Only ES004 and ES005 are retriggerable from the Business Events app. Events stuck in Processing or Transformed status for more than 15 minutes can be retriggered. Non-retriggerable errors (ES001–ES003) require fixing the root cause in the source system before resubmission.
Application Platform & Infrastructure	Integration	SAP Event Mesh & Cloud Events	EMC007	Event naming conventions	SAP Advanced Therapy Orchestration follows consistent event type naming conventions that indicate direct origin: outbound events from CGTO use sap/cgt/tmp/shipment/... or sap/cgt/order/... prefixes; inbound events from ERP use sap/cgt/erp/shipment and inbound events from external systems use sap/cgt/ext/.... SAP S/4HANA Cloud Events follow the pattern ce/sap/sci/cgto/... (inbound to S/4HANA) or ce/sap/s4/beh/... (outbound from S/4HANA). This

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					it straightforward to identify event direction and or from the event type alone.
Application Platform & Infrastructure	Integration	SAP Batch Release Hub Integration	BRH001	Multi-level batch release decisions	SAP Advanced Therapy Orchestration integrates w Batch Release Hub for Life Sciences (BRH) via SAP S/4HANA for multi-level batch release decision management. When an inspection lot is created in S/4HANA, it triggers a release decision item in BRH quality responsible user makes the release decision BRH, which updates the inspection lot in SAP S/4H and the SAP S/4HANA add-on fetches the decision and sends it to SAP Advanced Therapy Orchestration. Multiple release decisions for the same batch are created and displayed on the Release Decision History tab. Multi-level release is essential for CGT products that require quality decisions at multiple manufacturing stages. Requires SAP S/4HANA Life Sciences product add-on 2310. Introduced in release 2404.
Application Platform & Infrastructure	Integration	SAP Batch Release Hub Integration	BRH002	Biospecimen batch disposition	Batch release for biospecimen materials can occur in S/4HANA or an external system (such as SAP Batch Release Hub). When a usage decision is updated for an inspection lot, the disposition event is sent to SAP Advanced Therapy Orchestration to update the biospecimen shipment. The disposition event is triggered against the material document BAdI (changed from inspection lot update BAdI in release 2404) to ensure consistent processing. Usage decision code and description are included in the event payload (enhancement 2502).
Application Platform & Infrastructure	Integration	SAP Batch Release Hub Integration	BRH003	Quality inspection for STOs	Batch disposition from quality inspections at the receipt plant for stock transport orders is captured and sent to SAP Advanced Therapy Orchestration for processing. SAP S/4HANA add-on sends the STO Disposition List event to SAP Advanced Therapy Orchestration, and SAP Advanced Therapy Orchestration forwards the disposition information to the treatment center via the Send Disposition Information Update outbound event. This covers both intermediate product and finished product transit leg shipments. Introduced in release 2502.
Application Platform & Infrastructure	Integration	SAP Batch Release Hub Integration	BRH004	Batch genealogy	In the Monitor Chain of Identity (COI) app, product events can display batch genealogy information retrieved in real-time from the Global Batch Trace (GBT) system. This requires an HTTP destination configured for the system (using the DEFAULT_GBT rule provider) and the Batch Genealogy Information attribute enabled in the Event Profile for the Production End event. The genealogy provides upstream/downstream traceability of how batches relate to each other across manufacturing.
Application Platform & Infrastructure	Integration	SAP Batch Release Hub Integration	BRH005	Release Decision feature flag	Multi-level batch release decision history is controlled by the feature flag ft_release_decision, which must be enabled via an SAP support case. When enabled, the

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					Release Decision History is displayed in material ba details across the Manage Shipment (Biospecimen Manage Shipment (Finished Product), and Manage Processing Activity apps. The feature is not support non-batch-managed materials, and Release Decisi History is not available in the Manage Inbound Log Manage Outbound Logistics apps.
Application Platform & Infrastructure	Integration	Data & Document Integration (DDI)	DDI001	DDI add-on architecture	The Data and Document Integration for SAP Cell at Therapy Orchestration (DDI) is a customer-deploye Docker/Kubernetes add-on that acts as a stateless conduit between SAP Advanced Therapy Orchestra and customer-controlled data, document, and lab systems. SAP Advanced Therapy Orchestration doe store protected health information (PHI) – it only v PHI from customer-controlled storage via this add-add-on provides access and display layer functiona with PHI stored in customer-controlled AWS S3 Ob Store. This architecture reflects a key design princi personalized medicine: because each treatment or linked to an identifiable patient, PHI must remain u customer's data sovereignty rather than in a share database. The last feature release was version 240
Application Platform & Infrastructure	Integration	Data & Document Integration (DDI)	DDI002	Document Vault API integration	Introduced in release 2404, this capability allows integration with external document vault APIs for s uploading, and viewing documents, replacing the p Google Cloud Services bucket-only approach. Cust can switch between different vault APIs based on b scenario using VaultInfo.json routing in the Config Configuration requires VaultInfo.json (routing rules Data Vault ID, Document Vault ID, and Label Type) a Document-api-secrets.json (credentials). Supports sizes up to 500MB.
Application Platform & Infrastructure	Integration	Data & Document Integration (DDI)	DDI003	PHI access architecture	SAP Advanced Therapy Orchestration accesses PH BTP destination (DEFAULT_CONNECTIVITY_URL) connects to the DDI add-on. The add-on supports t integration scenarios: with satellite endpoints enab (multiple PHI data vaults routed by Patient Therapy Number) and without (single vault). Three authenti methods are available: Basic, API Key, and Certificate Read Access Logging records who accessed what F and when, writing to the SAP Audit Log for complia
Application Platform & Infrastructure	Integration	Data & Document Integration (DDI)	DDI004	End-to-end encryption	Optional encryption of the request payload exchan between SAP Advanced Therapy Orchestration and DDI add-on, using the SAP Credential Store service encryption uses RSA-OAEP-256 + A256GCM algori This provides an additional layer of security for sen data in transit beyond the standard HTTPS transp encryption. Introduced in release 2404.
Application Platform &	Integration	Data & Document	DDI005	Session management	Optional capability to make the DDI add-on sessior explicitly dependent on the SAP Advanced Therapy

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Infrastructure		Integration (DDI)			Orchestration session, ensuring that the add-on se lifecycle is tied to the authenticated user session in Advanced Therapy Orchestration. This prevents sta orphaned sessions in the add-on. Introduced in rel 2404.
Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX001	Custom field creation	Customers can add custom fields to the biospecimen shipment entity (Shipment and ShipmentMaterial) the SAP Cloud Application Programming Model (CAP). Extension fields support five data types: Boolean, Integer, Decimal, String, and DateTime. A maximum of 20 custom fields can be added per entity. New entities cannot be created – only field-level extensions are supported. This enables CGT manufacturers to capture therapy-specific data – such as custom collection site identifiers, donor attributes, or proprietary quality parameters – directly on the biospecimen shipment without product modification.
Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX002	Field naming convention	All extension field names must start with one of the reserved prefixes: Y_, YY_, Z_, or ZZ_. This naming convention ensures clear separation between SAP-delivered fields and customer extensions, preventing naming conflicts during upgrades. The convention is enforced during the cds push deployment step and applies to both database model and UI annotation definitions.
Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX003	Extension development workflow	Extension development follows the SAP CAP lifecycle: cds init (create project) -> write extension (data model annotations + i18n labels) -> cds login (authenticate) -> cds pull (get current model) -> cds build (compile) -> cds push (deploy and activate on cloud). The ExtensionDeveloper role is required for deployment. Extension projects must contain language-specific Attribute Labels CSV files for localization of field labels in the UI.
Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX004	Event interface for extension fields	Custom extension fields are included in both inbound and outbound event payloads. Supported events include Biospecimen Shipment Create, Biospecimen Shipment Update, Biospecimen Collection Shipment Create, Biospecimen Collection Shipment Update, and Purchase Order Creation. This allows external systems to both send and receive customer-specific data through the standard event interfaces without custom integration.
Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX005	E-signature for extension fields	When extension fields are annotated with @cgt.signature.enabled, any update to those fields follows the standard e-signature validation workflow. This ensures that customer-specific data extensions are held to the same GxP compliance standards as SAP-delivered data. Currently supported on Biospecimen Shipment and Finished Product Shipment apps. Introduced in release 2404.

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Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX006	Change log for extension fields	Extension fields annotated with @CGT.changelog.e automatically generate change log entries whenever they are created or updated, whether the change occurs manually through the UI or via an inbound event. Multilingual language label support via CSV files ensures that change log entries display field names in the user's language. Currently supported for Biospecimen Shipments and Collection Shipments. Introduced in release 2404.
Application Platform & Infrastructure	Extensibility	Field Extensibility	FEX007	Visibility control for extension fields	The visibility of extension fields can be controlled using field control profiles via virtual _fc fields and the Import Extension Attributes button in the Configure Field Control Profiles app. This allows administrators to show or hide customer-specific fields per order type or shipment without modifying the extension code. Currently supported only for Finished Product Shipment app and Biospecimen Shipment entity and localization of extension fields in the profiles app are not yet supported.
Application Platform & Infrastructure	Extensibility	Custom Texts	CTX001	Configure Custom Text Types app	A dedicated configuration app allows administrators to define custom text types, each with a unique ID and a description. Text types can be searched, filtered, created, updated, activated, and deactivated. SAP recommends using a unique text ID per business entity to avoid duplication when custom texts are included in outbound events to SAP S/4HANA. Roles CGT_GeneralData_Config_Read_Srv and CGT_GeneralData_Config_Write_Srv provide access. Introduced in release 2502.
Application Platform & Infrastructure	Extensibility	Custom Texts	CTX002	Additional Information tabs	A new "Additional Information" tab (or "Additional Information" for treatment orders) is available across Manage Orders and all Manage Shipments apps (Biospecimen, Finished Product Final Leg, Finished Product Transit Leg, Intermediate Product, Biospecimen Kit). Users can create, read, update, and delete custom text entries on each entity. For treatment orders, the previous "Order Texts" tab was renamed to "Additional Order Information." Introduced in release 2502.
Application Platform & Infrastructure	Extensibility	Custom Texts	CTX003	Multi-language support	Custom texts support 35+ languages, enabling global deployments where the same custom text type can have values in different languages for different users or regions. Language codes follow standard ISO conventions. This makes custom texts suitable for regulatory annotations, manufacturing instructions, or any structured metadata that needs to be maintained in multiple languages for international therapy programs.
Application Platform & Infrastructure	Extensibility	Custom Texts	CTX004	Header and material batch level texts	Custom texts are supported at the header level for entities (treatment orders, all shipment types). For biospecimen shipments, custom texts are also supported at the material batch level, providing granular metadata attachment to individual batches within a shipment. This two-level structure enables both shipment-wide

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					annotations and batch-specific notes (e.g., batch-handling instructions or quality observations).
Application Platform & Infrastructure	Extensibility	Custom Texts	CTX005	Inbound event support	Custom text arrays have been added to inbound events for all major entity types: Treatment Order Creation/Update, Biospecimen Shipment Create/Update, Finished Product Shipment Create/Update, Biospecimen Kit Shipment Create/Update, and Intermediate Shipment/Finished Product Transit Leg Update. External systems can read custom texts when creating or updating entities, with texts automatically stored and displayed in the corresponding Additional Information tab.
Application Platform & Infrastructure	Extensibility	Custom Texts	CTX006	Outbound event support to SAP S/4HANA	Custom texts are included in outbound events to SAP S/4HANA: Process Order Creation, Purchase Order Creation, Sales Order Creation, and Stock Transport Request Creation. A dedicated Custom Text Update event (sap/cgt/tmp/shipment/custom_text/update) is triggered when custom texts are changed or deleted, enabling real-time synchronization. In SAP S/4HANA, custom texts are viewable via the standard Texts tab in SAP GUI. A corresponding SAP S/4HANA inbound event (ce/sap/sci/cgto/CGTOText/Maintain/v1) processes text data.
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX001	Status Extension event	The PPF event Status_Extension (Workflow Event: Entity Update Request) enables custom status transitions via inbound events using the event type sap/cgt/tmp/entity/status_transition/update. This enables external systems to trigger status changes on SAP Advanced Therapy Orchestration entities without direct API access, extending the standard status management workflow with customer-specific business logic. Supported entities include: Treatment Orders, Biospecimen Shipments/Material/Subunits, Finished Product Shipments/Material/Subunits, Processing Activities/Tasks. This is valuable in CGT orchestration where external systems such as ordering platforms, LIMS, or MES advance entities to the next status based on events occurring outside SAP Advanced Therapy Orchestration.
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX002	Workflow trigger on shipment	Synchronous outbound API calls from SAP Advanced Therapy Orchestration to SAP Build Process Automation (BPA) workflow instances are triggered when a shipment business object is created or modified at the header or sub-entity level. All five shipment types are supported: Biospecimen Kit, Biospecimen, Intermediate, Finished Product Transit Leg, and Finished Product Final Leg. These calls are visible in the Monitor Business Events application as OUTBOUND_SYNCHRONOUS, with both the sent payload and received response viewable. Requires PPF events Notify_All_Shipment_Create and Notify_All_Shipment_Change plus corresponding rules in the Configure Rules app. Introduced in release 250, Workflow notifications are now triggered for any shipment.

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					to custom milestones within Shipments (including BS, BSK, IS, FSIL, FSFL), whether the update occur or inbound event/API
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX003	CPI iFlow extension points	All existing CPI integration flows (iFlows) in the SAI Advanced Therapy Orchestration integration pack have been enhanced with both pre-exit and post-exit extension points. Pre-exit extensions allow customers to modify message payload before standard mapping and export. Post-exit extensions allow both message modification and rerouting of outbound calls to different receivers via custom routing logic, enabling customers to redirect integration flows without modifying the standard iFlow. Customers must re-import iFlows from SAP Business Accelerator Hub to receive the extension points. Introduced in release 2506.
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX004	SAP Build Process Automation support	SAP Advanced Therapy Orchestration supports SAP Process Automation as the platform for managing workflows and business rules. This service enables customers to define approval workflows, create and configure business rules, and implement custom process automation. SAP Build Process Automation replaces legacy SAP Workflow Management service as the recommended platform starting with release 2409. The legacy service continues to be supported for backward compatibility.
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX005	Feature flags for controlled enablement	Certain new features introduced in SAP Advanced Therapy Orchestration releases are delivered in a disabled state behind feature flags to minimize impact on customers running the product in validated GxP environments. This ensures that existing workflows remain unaffected when a new release is deployed. When customers are ready to adopt a new feature, they raise a support case with the corresponding flag enabled, allowing them to test and validate the feature in their pre-production environment before going live. Ten feature flags are documented as of the latest release, covering capabilities such as order cloning, scheduling, material determination and release decisions.
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX006	Workflow trigger on treatment order	Synchronous outbound API calls from SAP Advanced Therapy Orchestration to SAP Build Process Automation (BPA) workflow instances are triggered when a treatment order business object is modified for custom milestones.
Application Platform & Infrastructure	Extensibility	Process Extensibility	PEX007	Workflow trigger on processing activity	Synchronous outbound API calls from SAP Advanced Therapy Orchestration to SAP Build Process Automation (BPA) workflow instances are triggered when a processing activity business object is modified for custom milestones.
Cross Applications	Treatment Management	Treatment Order Management	TOM001	Treatment order lifecycle	SAP Advanced Therapy Orchestration manages the complete treatment order lifecycle from creation to closure. Orders are typically initiated in an external onboarding or treatment request application and

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					submitted to SAP Advanced Therapy Orchestration Treatment Order Creation inbound event. The lifecycle encompasses validation, data exchange with ERP and external systems, fulfillment of all steps defined in version, and eventual closure. All other SAP Advanced Therapy Orchestration features are designed to support this central orchestration process. Each treatment represents a single patient's therapy journey – from moment cells are collected through manufacturing release, and delivery of the finished product for infusion making it the primary unit of work around which the CGT supply chain is organized.
Cross Applications	Treatment Management	Treatment Order Management	TOM002	Manage Orders app	The primary user interface for treatment order management providing a comprehensive view of all treatment orders with their associated shipments, processing activities, tasks, and exceptions. The app supports viewing and editing general information, patient details, approval information, and related entities. Field visibility is configurable per order type via field control profiles. The app has been progressively enhanced with IS shipment tabs (2404), FS-IL shipment columns and flow version visibility (2409), Related Orders section (2409), new (2502), analytical page (2506), and Treatment Order Summary (2511). This also offers a section "Milestones" and displays both standard and custom milestones.
Cross Applications	Treatment Management	Treatment Order Management	TOM003	Treatment order creation and update	Treatment orders are created via the Treatment Order Creation inbound event, which carries patient details, therapy information, COI ID, and order-specific data. The Treatment Order Update outbound event notifies external systems of changes. New fields added over time include Route ID, Donor ID, Subject Status (configurable dropdown), and Study ID (2502), plus Clinical Trial Number, Patient Therapy Number, Protocol ID, and Randomization Variable from earlier releases. Donor ID is not editable in the Manage Orders app and displayed in the shipment and processing activity apps (2506).
Cross Applications	Treatment Management	Treatment Order Management	TOM004	Treatment order approval	Treatment orders can require approval before processing proceeds. The approval workflow is managed via SAP Process Automation, with the Treatment Order Approval outbound event notifying external systems of the approval request. The Approval Information API (POST /UpdateApprovalInfo) allows external workflow systems to update approval details, status, and metadata based on the workflow task ID. Approval details include a "Done By" field showing who made the approval decision.
Cross Applications	Treatment Management	Treatment Order Management	TOM005	Treatment order cancellation	Orders can be cancelled via the Treatment Order Cancellation inbound event, which triggers a cancellation workflow. The Treatment Order Cancellation Response outbound event confirms the cancellation status back to the requesting system. Cancellation requires appropriate authorization roles.

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					(CGT_TreatmentOrder_Request_Cancel_Ac) and for the same request -> approve/reject pattern used across entity types. In CGT, cancellations occur when a patient condition changes, a therapy is discontinued, or manufacturing fails – the dual-control pattern ensures these consequential decisions are deliberate and documented.
Cross Applications	Treatment Management	Treatment Order Management	TOM006	Follow-on orders (clone)	When a flow version is confirmed and cannot be changed, a follow-on treatment order can be created with the same COI ID but a different flow version or therapy. This is the primary mechanism for handling therapy reclassification (e.g., commercial to clinical) and flow version change. The clone introduces an Active/Inactive state across all orders – the parent order and all its sub-entities are marked Inactive, while the follow-on order is created as Active. All inbound events are automatically routed to the active order. Controlled by feature flag ft_enable_order_clone. Introduced in release 2409. When a follow-on order is created, all existing scheduling requests linked to the inactive (original) order are now automatically marked Obsolete or Partially Obsolete depending on milestone status, and all future slots are released. Completed process entities (including milestones and planned orders) are now accurately cloned onto the new active Treatment Order.
Cross Applications	Treatment Management	Treatment Order Management	TOM007	Copy Control for cloning	Copy Control profiles configure which optional data sections are copied when creating a follow-on order. Profile types exist: Treatment Order (Business Part, Order Texts, Order Dates, Documents, Approval Documents), Shipment (Documents, Logistics, Approval Info, ERP Data) and Processing Activity (Documents, ERP Data). Documents are shown on the UI but users can override at clone time. Configure Order Types defines which follow-on order types are valid per therapy, with activation/deactivation and default marking. Introduced in release 2409.
Cross Applications	Treatment Management	Treatment Order Management	TOM008	Change Key Attributes event	A powerful inbound event enables updating critical treatment order attributes with cascading ERP document management. Supported attributes include: Plant (validates material+plant, cancels/recreates Process Order), Biospecimen Material (validates, creates new/deletes old, cancels/recreates Purchase Order), Finished Product Material (validates allocation status, creates new/deletes old, cancels/recreates Sales Contract) and Biospecimen Delivery/FP Pickup Location (updates with org-plant mismatch warnings). Any combination of inputs is supported in a single event. Controlled by feature flag ft_enable_order_clone. Introduced in release 2409.
Cross Applications	Treatment Management	Treatment Order Management	TOM009	Create Subsequent entities	The Create Subsequent button in the Manage Order screen allows creating additional processing activities or shipments at the same flow step, copying data from the selected parent entity. For processing activities, it creates

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					general data, flow version details, PA locations, and materials (CMO type only). For shipments, it copies version details, locations (from/to organization, address contacts), and materials. This enables handling exceptions scenarios where additional entities are needed at a stage. Introduced in release 2409.
Cross Applications	Treatment Management	Treatment Order Management	TOM010	Patient Details Update event	The Patient Details Update inbound event (sap/cgt/order/patient/updateNotification) enables external systems to update patient-specific information on an existing treatment order after creation. This supports scenarios where patient data is corrected or supplemented after the initial order submission. The Clinical Trial Subject ID and Patient ID are also sent to SAP Advanced Therapy Orchestration to SAP S/4HANA biospecimen and finished product shipments via the Purchase Order Inherit Order Change event (2502)
Cross Applications	Treatment Management	Treatment Order Management	TOM011	Treatment Order Information APIs	Two versions of the Treatment Order Information APIs provide read access: V2 (GET /TreatmentOrderAPI) retrieves all orders with COI ID, Treatment Center ID, Therapy ID, and Patient Therapy Number, with extended \$expand options added in 2502 (pnConfirmSchedule, shipConfirmSchedules, planningObjectVersions, planningDetails, bkShipments, isShipments, document and 20+ new fields. The Treatment Order Summary (GET /TreatmentOrderSummaryAPIView) provides a consolidated view of milestone dates, processing a material quantities, courier details, and ERP document details. Introduced in release 2511.
Cross Applications	Treatment Management	Treatment Order Management	TOM012	Send Treatment Order Related Entities	On flow version confirmation, the outbound event sap/cgt/order/treatment_order/related_entities is triggered, sending all entity IDs (shipments, processing activities, related orders) to external systems. This enables ordering platforms and external systems to synchronize their data model with the confirmed flow version. The event was enhanced in 2511 to include a Scheduling section with milestones, scheduling request ID, product ID, and success/error messages. Introduced in release 2502.
Cross Applications	Treatment Management	Treatment Order Management	TOM013	Receive Milestone updates	Introduced support for receiving and updating both standard and custom milestones in CGTO from external systems via the new sap/cgt/ext/shipment/milestone/update event.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP002	Flow version confirmation	A one-time action that locks the flow version for a treatment order and creates all associated entities (shipments, processing activities) based on the flow definitions. During confirmation, the system determines intermediate product materials via business rules, validates materials against source and destination, creates shipments (including IS, FS-IL if defined), and determines plants and addresses for parallel flows.

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					confirmed, the flow version cannot be changed – o follow-on order can use a different version. This immutability is a deliberate GxP design choice – it an auditable baseline for the therapy supply chain validated before execution began.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP003	Shipment type support in flow definitions	Flow definitions support five shipment types: Biospecimen (BS), Biospecimen Kit (BKS), Intermediate Product (IP), Finished Product Transit Leg (FS-IL, for inter-plant) – added in 2404, and Finished Product Final Leg (FS-FL). The flow step structure can model simple single-plant scenarios, multi-plant scenarios with intermediate shipments, and transit leg scenarios where finished product passes through intermediate drop-off center before final delivery.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP004	Co-product and parallel flows	Processing activities can yield a main product plus more co-products, each with its own parallel flow branch. The Additional Step ID field (introduced in 2502) allows product and co-product flows are processed in parallel. In complex scenarios, a single biospecimen processing activity can branch into a direct FS-FL delivery for the main product and a multi-plant processing chain with transit legs for the co-product. The Requested Flow Step ID on FP Final Leg shipments links each shipment to the correct parallel flow branch.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP005	Milestone management	Milestones are auto-generated from flow steps and the scheduling algorithm. Thirteen standard milestone types cover all entity types (BSCOLL, BSPICKUP, BSKPICKUP, BSKDEL, PASTART, PAEND, ISPICKUP, FSILPICKUP, FSILDEL, FSFLPICKUP, FSFLDEL). Each milestone is configured with a Next Milestone ID, Lead Time (days), Mark Not Applicable flag, and Auto Block Subsequent Slots option. Lead times drive slot scheduling. Given a preferred pickup date, the system calculates downstream dates. Regeneration preserves existing milestones and generates new ones only for changed steps. Milestone-driven scheduling is critical for CC therapies where manufacturing slot availability, co-product transit times, and product viability windows must be maintained – a miscalculated lead time between biospecimen pickup and processing start could result in cells arriving past viability window.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP006	ERP document type configuration	Flow version definitions configure which ERP document type is created per shipment on schedule confirmation. Defaults are: Purchase Order for biospecimen shipment, Sales Order for finished product final leg and biospecimen kit shipments, and Stock Transport Order for intermediate product shipments. For transit leg shipments, the ERP document type can be configured as either Stock Transport Order or Sales Order. This determines the

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					document lifecycle that flows through SAP S/4HANA. Introduced in release 2502.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP007	Process flow visualization	The Manage Orders app displays a tabular process showing entities (Level 1) and process steps (Level Status, Previous Step, and Next Step columns as navigation links, plus a View Details popup displaying entity attributes. The attribute display is configurable. Three new field control profiles: Processing Activity Flow, Shipment Process Flow, and Treatment Order Process Flow, associated via Configure Profile Group. Graphical view of the process flow is not yet supported. Introduced in release 2506.
Cross Applications	Treatment Management	Flow Version & Process Flow	FVP008	Lead times between milestones	The Manage Flow Definitions app includes a Lead Time tab supporting generation and configuration of lead times between standard milestones for key flow steps. Users can configure and maintain milestones directly in the app with system validation on save. On regeneration, existing milestones are preserved and new ones are generated for changed flow steps. This enables fine-tuning of scheduling algorithm per therapy and manufacturing network. Enhanced in release 2511.
Cross Applications	Treatment Management	Order Scheduling & Confirmation	OSC001	Schedule confirmation	The Confirm Schedules app validates planned dates and triggers ERP document creation in SAP S/4HANA. For schedule confirmation, each shipment and processing activity receives its corresponding ERP document: Purchase Order (BS), Sales Order (FS-FL, BKS), Stock Transport Order (FS-IL), and Process/Production Order (PA). All shipments are displayed in a single table differentiated by shipment type and sorted by flow step. Individual confirmation for BS, FP, or PA is supported as standalone requests in SAP S/4HANA (2404), and header-level fields were renamed: "Schedule Last Confirmed By/On" with item-level timestamps added.
Cross Applications	Treatment Management	Order Scheduling & Confirmation	OSC002	Automatic schedule confirmation	Schedules can be auto-confirmed in two scenarios: for new orders when the flow version is confirmed (requires PPF configuration and validated planned dates), or for existing orders with a confirmed flow version when subsequent shipments or processing activities are created via events with the autoConfirmSchedule flag set to true. This flag has been added to the Biospecimen Shipment Creation, Finished Product Shipment Creation (Fin Prod), and Processing Activity Create events. Introduced in release 2502.
Cross Applications	Treatment Management	Order Scheduling & Confirmation	OSC003	Business partner determination on confirmation	During schedule confirmation, the system checks for minimum required business partners per shipment. For Vendor/Supplier for biospecimen shipments, Bill To and Ship To for finished product transit leg shipment. If required partners are missing, business rules (BRF470) attempt to determine them automatically. A single match populates the Business Partners tab; multiple matches are listed in the Business Partners tab. Introduced in release 2502.

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					matches result in unsuccessful determination. The schedule is confirmed regardless of determination success, ensuring the process is not blocked. Enh 2409 for FS-FL, extended in 2502 for BS and FS-IL
Cross Applications	Treatment Management	Order Scheduling & Confirmation	OSC004	Schedule approval and hard booking	The Approve Schedule button in the Confirm Schedule app (controlled by feature flag ft_support_scheduling) triggers hard booking of manufacturing slots when proposed schedule dates are accepted. Approving proposed dates converts soft-booked slots to permanent hard-booked reservations. If dates are changed, the system re-checks slot availability and generates a new proposal. Hard booking is essential for CGT where production capacity is extremely limited and a double-booked slot could delay a patient's therapy. The app triggers the outbound event with a Scheduling section containing milestones, scheduling request ID, proposed dates, and success/error messages. Introduced in release 2502.
Cross Applications	Treatment Management	Treatment Order Analytics & Summary	TAS001	Analytical page with graphical filters	The Manage Orders app includes an analytical page with graphical filters for Therapy, Country, Treatment Center, and Order Status. A graphical report can be displayed based on treatment order header attributes, with configurable X and Y axes. The default report shows Number of Orders by Status on the X axis and Country on the Y axis. Additional fields can be added to person filters and columns, including Donor ID, Subject Status, Route ID, Protocol ID, Clinical Randomization Variable, Study ID, Flow Version, Open Tasks/Exceptions, and Document/Data Vault IDs. This gives CGT program managers portfolio-level visibility into treatment order throughput and bottlenecks across their manufacturing network. Introduced in release 2506.
Cross Applications	Treatment Management	Treatment Order Analytics & Summary	TAS002	Treatment Order Summary view	The Manage Treatment Orders app includes a Treatment Order Summary view that consolidates key information from across the treatment order into a single screen. The summary displays Milestone Dates (planned/actual and delivery for shipments), Processing Activities (planned/actual processing dates and plant), Material Quantities (dosage, requested), Courier Details (Country and Waybill), and ERP Details (Sales Order, Purchase Order, Process Order, Stock Transport Order number and status). This reduces the need to navigate between multiple apps for order monitoring and approval. Introduced in release 2511.
Cross Applications	Treatment Management	Treatment Order Analytics & Summary	TAS003	Treatment Order Summary API	A dedicated REST API (GET /TreatmentOrderSummaryAPIView) provides programmatic access to the same consolidated summary data available in the UI. The API returns shipment milestones, processing activity details, material quantities, courier information, and ERP document status for a single order or all orders. This enables external dashboard reporting tools, and ordering platforms to display

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					summary information without querying multiple endpoints. Introduced in release 2511.
Cross Applications	Treatment Management	Treatment Order Analytics & Summary	TAS004	Treatment Order Information API	The Treatment Order Information V2 API (GET /TreatmentOrderAPIView) provides comprehensive access to treatment order data with extensive filter navigation capabilities. Release 2502 significantly enhanced the API with new \$expand options (pnConfirmSchedules, shipConfirmSchedules, planningObjectVersions, planningDetails, bkShipments, documents) and 20+ new fields including scheduling details, custom texts, and shipment type previously exposed.
Cross Applications	Treatment Management	Treatment Order Analytics & Summary	TAS005	Order Overview app	The Order Overview app is a standalone analytical dashboard providing a portfolio-level view of all treatment orders in the system. It presents five interactive tile views: (1) Orders By Overall Status – donut chart of order counts by status with click-through to the order list, (2) Orders Waiting For Flow Version Confirmation – subset of pending flow version confirmation with navigation to Confirm Flow Version app, (3) Orders By Approval Status – donut charts for create, change, and cancel approval statuses (Approved, Rejected, Sent For Approval), (4) Orders By Actual vs Planned Dates – delay tracking across six milestones (Biospecimen Pickup, Biospecimen Delivery, Processing Start, Processing Completion, Finished Product Pickup, Finished Product Delivery), (5) Orders By Processing Status – status breakdown across 11 processing steps from Incoming Biospecimen through Allocate Batches and Confirm Pickup. Use filters by Therapy, Treatment Center, and other attributes to slice the data, then click through any tile to drill down to the underlying order list. Access requires the CGT_OrderOverviewPage_Read_Srv role.
Cross Applications	Control Tower	Task Management	TSK001	Manage Tasks app	The primary user interface for creating, assigning, tracking tasks across the treatment order lifecycle. Tasks are linked to reference objects (treatment orders, shipments, or processing activities) via the reference object ID, providing context-specific tracking. The app supports searching, filtering, and viewing all tasks assigned to the same reference object via the Related Tasks view. Authorization is controlled by three roles: CGT_Task_App (tile access), CGT_Task_Read_Srv (view) and CGT_Task_Write_Srv (create/edit).
Cross Applications	Control Tower	Task Management	TSK002	Task properties and tracking	Each task carries a comprehensive set of properties: Priority (Low, Medium, High), Status (New, In Progress, Hold, Resolved, Cancelled), Assignee (selected from User Registry), Due Date (with alerts when exceeded), Completion percentage for progress tracking. Task support free-text Comments for collaboration and Document uploads for attaching supporting files. The combination of structured status tracking and

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					unstructured collaboration makes tasks suitable for routine follow-ups and complex investigations.
Cross Applications	Control Tower	Task Management	TSK003	To-do lists	Each task can contain a structured sub-task check called a To-do list, enabling decomposition of complex tasks into discrete action items. To-do items can be created individually or pasted in bulk from Microsoft making it efficient to create multi-step checklists. This is particularly useful for GxP-regulated processes where deviation resolution might require multiple verification steps that need to be tracked individually within a task.
Cross Applications	Control Tower	Task Management	TSK004	Manual and workflow-generated tasks	Tasks can be created manually by users in the Manage Tasks app or generated automatically via SAP Build Process Automation workflows triggered by PPF business events. Workflow-generated tasks enable automatic creation in response to specific business scenarios such as a shipment delay triggering a follow-up task for a logistics coordinator. The PPF configuration determines which business events trigger which workflows, allowing customers to tailor the automation to their operational needs.
Cross Applications	Control Tower	Exception Management	EXM001	Manage Exceptions app	The central user interface for viewing, managing, and resolving exceptions across the treatment order lifecycle. Exceptions are linked to reference objects (orders, shipments, or processing activities) and carry properties including Type (Technical or Business), Priority (Low, Medium, High), Status (New, In Progress, On Hold, Resolved, Cancelled), Assignee, Completion percentage, and Due Date. Authorization is controlled by three roles: CGT_Exception_App (tile access), CGT_Exception_Read_Srv (read), and CGT_Exception_Write_Srv (create/edit). In CGT operations, exceptions range from temperature excursions during shipment to COI mismatches at goods receipt, each potentially affecting a patient's unique therapy product and requiring documented investigation and resolution.
Cross Applications	Control Tower	Exception Management	EXM002	Five exception sources	SAP Advanced Therapy Orchestration supports five distinct exception sources, each representing a different trigger mechanism: Manually Created (by users in the app), Internally Generated (by SAP Advanced Therapy Orchestration business logic), Externally Generated (external systems via the Exception Creation event), AI Generated (via SAP AI Core models, introduced in 2505), and Rule Generated (via SAP Build Process Automation business rules, introduced in 2506). The Exception field is automatically populated based on how the exception was created, providing clear traceability to the exception origin.

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Cross Applications	Control Tower	Exception Management	EXM003	Automatic exception generation – blocked disposition	Exceptions are automatically created when batch or subunit disposition is blocked for biospecimen material, intermediate product materials, or processing activity yield materials. This ensures that quality-related blocked material batches are immediately surfaced to the responsible users without requiring manual monitoring. The exception is created with source “Internally Generated” and linked to the corresponding shipment processing activity. Introduced in release 2502.
Cross Applications	Control Tower	Exception Management	EXM004	Automatic exception generation – ERP document failures	Exceptions are automatically raised when a create, or cancel request for an ERP document (process order, sales order, or purchase order) fails in SAP S/4HANA. SAP catches integration failures that could otherwise go unnoticed, ensuring that users are alerted when the other side of a transaction does not complete successfully. The exception provides details about which document integration failed and the nature of the error.
Cross Applications	Control Tower	Exception Management	EXM005	Automatic exception generation – inbound event errors	When an inbound event generates an error or warning during processing, an exception is automatically created in the system. Three exceptions are excluded from this list: errors in grouped subsequent events (because the system stops processing after the first failure), incorrect event types (which generate errors in the Monitor Business Events app instead), and the Exception Creation event itself (to prevent recursive exception generation). Introduced in release 2506.
Cross Applications	Control Tower	Exception Management	EXM006	Automatic exception generation – shipment receipt failures	During shipment receipt, checks are performed to verify shipping conditions and confirm the Chain of Identity (COI) of the shipment. If either check fails, an exception is automatically generated with source “Internally Generated”, Priority “High”, Type “Error”, and Reporter “System”. This applies to biospecimen shipments, intermediate product shipments, and finished product transit leg shipments. Introduced in release 2506.
Cross Applications	Control Tower	Exception Management	EXM007	Automatic exception generation – shipment pickup failures	When a shipment pickup is completed, the system checks whether the serial number on the shipper matches the shipper serial number in the system, and whether the product is packed into the shipper identified by the serial number. If either check fails, an exception is automatically generated with High priority and Error Type “System”. This applies to intermediate product shipments, finished product transit leg shipments, and finished product leg shipments. Introduced in release 2506.
Cross Applications	Control Tower	Exception Management	EXM008	Automatic exception generation – processing termination	When the status of a processing activity is set to “Terminated”, an exception is automatically raised with source “Internally Generated”, Priority “High”, Type “Error”, Reporter “System”, and Exception Type “Functional”. This ensures that manufacturing terminations – which represent significant deviations from the planned process – are immediately escalated and tracked to the responsible users.

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					the exception management workflow. Introduced in 2506.
Cross Applications	Control Tower	Exception Management	EXM009	AI-generated exception determination	SAP Advanced Therapy Orchestration supports AI-exception determination using a “Bring Your Own Model” approach. A workflow built in SAP Build Process Automation retrieves specific order data and invokes customer-trained AI model hosted on SAP AI Core AI Launchpad. The AI model determines whether a scenario constitutes an exception and assigns its priority (High/Medium/Low) and type (Error/Warning). After determination, the workflow creates the exception via the Exception Creation event with source AI_GEN. An AI Recommendation field is displayed in the Manage Exceptions app for AI-generated exceptions. The Bring Your Own Model approach allows CGT manufacturers to train models using their specific therapy data, enabling detection of situations and exception scenarios that rule-based systems cannot capture. Introduced in release 2506.
Cross Applications	Control Tower	Exception Management	EXM010	Rule-generated exception determination	The same workflow pattern as AI determination, but uses SAP Build Process Automation business rules instead of AI models. A workflow retrieves specific order data, invokes configured business rules to evaluate whether the situation is an exception, and creates the exception via the Exception Creation event with source RULE_GEN. Event-driven and rule determination are supported for: Treatment Orders, Biospecimen Shipments, Biospecimen Kit Shipments, Intermediate Product Shipments, Finished Product Transit Leg and Final Leg Shipments, and Processing Activities. Introduced in release 2506.
Cross Applications	Control Tower	Exception Management	EXM011	Exception Creation inbound event	The Exception Creation event (sap/cgt/ext/operations/exception/create) enables external systems to create exceptions directly in SAP Advanced Therapy Orchestration. The event payload includes the exception source, priority, type, reference object, and optional fields such as AI Recommendation. This is the mechanism used by all automated exception sources (AI, rules, and external systems) and provides a standardized interface for integrating exception management with any external monitoring or alert system. Introduced in release 2502.
Cross Applications	Control Tower	Exception Management	EXM012	Exception analytics	The Manage Exceptions app includes an analytics capability with graphical filters for Start Date, Due Date, Priority, and Status. An exception analytics chart provides a Pie Chart and Chart-and-Table view toggles with a default bar graph showing exception source and message type. Users can customize the visualization via View By and Select buttons to analyze exception patterns by different dimensions. This enables trend identification and root cause analysis across the treatment order portfolio. Introduced in release 2502.

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Cross Applications	Control Tower	Exception Management	EXM013	Exception notification emails	Email notifications can be triggered through SAP A Notification service for SAP BTP when exceptions are created or updated. Notifications support all five event creation sources (Manual, Internal, External, AI, Rule) and manual status updates. Configuration requires two event mappings: Business Context Create_Exception and Event Notify_Exception_Create, and Business Context Update_Exception with Event Notify_Exception_Update. The SAP Alert Notification service integration must be set up in the customer's BTP subaccount. Introduced in release 2506.
Cross Applications	Document & Label Management	Document Management	DMG001	Document vault linking	SAP Advanced Therapy Orchestration links documents to all major entities: treatment orders, shipments (all types), processing activities, material batches, and collections. Documents are stored in customer-controlled storage (via the Data and Document Integration add-on) and referenced from SAP Advanced Therapy Orchestration. This architecture ensures that SAP Advanced Therapy Orchestration never stores documents directly – it maintains metadata and references while actual files reside in customer-managed vaults. In addition, these documents include batch documentation files, identity documents, COI certificates, and traceability reports – all of which may contain patient data and remain under the customer's regulatory control.
Cross Applications	Document & Label Management	Document Management	DMG002	Document upload and download	Users can upload new documents into the document vault and download existing documents from any SAP Advanced Therapy Orchestration app that supports document management. Upload uses multipart/form-data with metadata including requestId (Patient Therapy Number), COI ID, description, document type, file path/name, file type, entity ID, and user email. Supported file formats include PDF and Microsoft Word, with a maximum size of 500 MB (subject to the vault API's own limits).
Cross Applications	Document & Label Management	Document Management	DMG003	Document type configuration	Document types are configured in SAP Advanced Therapy Orchestration with an isPHIRelevant flag (Yes/No) that determines whether accessing the document requires a PHI role (CGT_PHI_Read_Srv). Standard document types include BDF (Batch Documentation File), IDM (Identifier Document), COI (COI Certificate), and TRACEABILITY (Traceability Report). Entity Status Profiles for documents use the PhiDocument entity with Object Types per document type and Status Type PDS, controlling the lifecycle of document statuses.
Cross Applications	Document & Label Management	Document Management	DMG004	Document invalidation	Documents can be invalidated by users with the CGT_Document_Actions_Srv role, marking them as no longer valid without physically deleting them from the vault. Invalidation requires configured reason code and approximately 18 entity/source combinations. Invalid documents remain visible in the document log but are no longer presented as active documents. This supports

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					requirements where document versioning and audit must be maintained even when a document is sup
Cross Applications	Document & Label Management	Document Management	DMG005	Document Metadata Update event	The inbound event sap/cgt/document/metadata/ enables external document management systems update metadata associated with documents store SAP Advanced Therapy Orchestration's document This supports scenarios where document properties as approval status, version, or classification) are m in an external system and need to be synchronized SAP Advanced Therapy Orchestration without re-uploading the actual document content.
Cross Applications	Document & Label Management	Document Management	DMG006	Document access logging	Read Access Logging (RAL) records every access to relevant and sensitive documents, capturing who accessed the data, when they accessed it, and what they accessed (document URL). RAL events are written to the SAP Audit Log with 90-day retention and can be retrieved via Audit Log Retrieval API for long-term archiving. A SAP Document Read Access Log in the DDI add-on provides additional document-level access tracking.
Cross Applications	Document & Label Management	Document Management	DMG007	Document authorization model	Document access is controlled through a layered role model: CGT_Document_Read_Srv (view non-PHI documents), CGT_Document_Write_Srv (upload documents), CGT_Document_Actions_Srv (invalidate documents), and CGT_PHI_Read_Srv (view PHI-related documents). A user viewing PHI documents requires CGT_Document_Read_Srv and CGT_PHI_Read_Srv. Document change logs are accessible via CGT_ChangeLog_Read_Srv. This granular model ensures that document access follows the principle of least privilege.
Cross Applications	Document & Label Management	Label Management	LBM001	Label type configuration	SAP Advanced Therapy Orchestration supports four standard label types: BH (Biospecimen Hangtag), FP (Finished Product), SP (Secondary Packaging), and FPL (Finished Product Short Label). Labels are a critical part of Identity element in autologous CGT – the biospecimen hangtag accompanies patient cells from collection to manufacturing, and finished product labels must clearly identify the recipient to prevent mix-ups. Label types are configured in the system and linked to entity status profiles that govern the label lifecycle with statuses: Not Printed (NPR), Printed (PR), In Process (INPR), Rejected (RPR), and Invalidated (INV). Status transitions define allowed lifecycle paths (e.g., NPR->PR, PR->RPR/INV).
Cross Applications	Document & Label Management	Label Management	LBM002	Label template mapping	Labels are matched to templates via a five-attribute mapping: Therapy + Country/Region + Language + Label Type -> Label Template + Label Profile. The "*" wildcard can be selected for Country/Region to use a single template across all countries. Label template determination can also be automated via a Business Decision (one of two decision types exclusive to the

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					Business Decisions approach), which determines the template and language based on therapyId, labelType, countryCode, plantId, and materialId.
Cross Applications	Document & Label Management	Label Management	LBM003	Label printing apps	Three dedicated apps support label printing operations: Print Biospecimen Hangtag Labels (CGT_BHLabelPrinter_App), Print Finished Product/Secondary Packaging Labels (CGT_FPSPLabelPrinter_App), and Manage Labels (CGT_ManageLabels_App). Label print operations (generate, reprint, invalidate) require the CGT_LabelPrint_Actions_Srv role. Read access uses CGT_LabelPrint_Read_Srv with labelType attribute filtering, and write access uses CGT_LabelPrint_Write_Srv.
Cross Applications	Document & Label Management	Label Management	LBM004	Label print payload	When a label print request is sent to the external label management service via the DDI add-on, a rich payload is assembled including: label context (entity, template mode), material subunit details, shipment details (addresses, courier information, materials), patient details (name, date of birth, weight), processing node details, treatment order details (COI ID, therapy, country, treatment center) and optional checksum. This comprehensive payload enables the external label management system to produce fully populated labels without additional data lookups.
Cross Applications	Document & Label Management	Label Management	LBM005	Label printing setup	Label printing requires a six-step configuration: (1) user statuses mapped to NPR/PR/INPR/RPR/INVP, (2) create status profiles with transitions, (3) map entity status profiles for LabelDocument entity per label type, (4) configure reason codes for Reprint and Invalidate actions, (5) configure BTP destinations for the DDI add-on (DEFAULT_CONNECTIVITY_URL and Label Template Validation), and (6) set up a background job that reprints labels stuck in "In Process" status (default every 24 hours, customizable via SAP support ticket).
Cross Applications	Document & Label Management	Label Management	LBM006	Label e-signature logs	Label generation, reprinting, and invalidation actions are tracked through e-signature logs, ensuring that every operation is attributable to a specific authenticated user. These logs are viewable within the label printing application and contribute to the overall GxP audit trail. Combined with status lifecycle management, this provides full traceability of every label produced, reprinted, or invalidated throughout the manufacturing process.
Cross Applications	Document & Label Management	Certificate & Traceability Report Generation	CTR001	COI certificate generation	COI certificates contain patient, biospecimen, and shipment information and are generated for Finished Product Leg (FS-FL) shipments at the shipment or batch level. They serve as formal proof that the correct patient's cell was used to manufacture the therapy – a regulatory requirement for autologous CGT. Generation requires a configured certificate template (mapped to the organization via Configure Profile Groups), a valid e-signature, a

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					Data and Document Integration add-on. The CGT_CoiCoc_Document_Action_Srv role authorize certificate generation from the Manage Orders app
Cross Applications	Document & Label Management	Certificate & Traceability Report Generation	CTR002	Traceability report generation	The Generate Traceability Report button on the Documents tab of Manage Orders produces a PDF of all COI and COC events for a treatment order. Re use configurable templates (.xdp format) with SAP and watermark, and can be saved to the app and downloaded as PDF. Multiple report versions can be generated for the same order. Requires SAP BTP Advanced Form Service and SAP BTP Object Store (AWS S3 compatible). Only active orders are supported. Introduced in release 2502.
Cross Applications	Document & Label Management	Certificate & Traceability Report Generation	CTR003	Profile-driven report content	Starting with release 2506, the traceability report content is controlled by COI and COC event profile configuration. Events and attributes disabled in the profiles are excluded from the report. Configurable display/print combinations include: all COI events with one COC root event, all events with one COI root event, all COI and COC event selected events only. If no event profiles are configured, the order type, all events and attributes are shown backward compatibility. Common attributes (Event Description, Date/Time, Performed By, Material, Batch, etc.) cannot be controlled via profiles.
Cross Applications	Document & Label Management	Certificate & Traceability Report Generation	CTR004	Certificate template configuration	Certificate and traceability report templates are configured in the Configure Profile Groups app, link template files to specific order types. Templates must be in .xdp format (Adobe Experience Manager forms) and stored via the SAP BTP Object Store. The template configuration supports logo customization and watermark settings, allowing customers to brand their compliance documents according to their corporate and regulatory requirements.
Cross Applications	Document & Label Management	PHI & Consent Management	PCM001	PHI viewing architecture	SAP Advanced Therapy Orchestration provides authorization-based access to protected health information (PHI) stored in the customer's data vault as a fundamental architectural principle: SAP Advanced Therapy Orchestration does not store PHI. The DDI acts as a stateless conduit between SAP Advanced Therapy Orchestration and customer-controlled storage (AWS S3). PHI is displayed in the UI only when a user with the CGT_PHI_Read_Srv role accesses the relevant data and the data is retrieved in real-time from the customer data vault without caching or persistence in SAP Advanced Therapy Orchestration. This zero-storage PHI architecture is particularly important for personalized therapies as each treatment order is linked to an identified patient, unlike conventional drug manufacturing where product level data rarely contains patient identifiers.

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
Cross Applications	Document & Label Management	PHI & Consent Management	PCM002	Multiple data vault routing	When the satellite endpoints feature is enabled (via <code>ft_satellite_create</code> feature flag), different treatment or therapies can be routed to different PHI data vaults. VaultInfo.json configuration in the DDI add-on maps Vault IDs, Document Vault IDs, and Label Types to specific API endpoints. The Patient Therapy Number is sent to the DDI add-on for vault routing when satellite endpoints are enabled. This supports organizations that segregate patient data across different storage systems for regulatory or organizational reasons.
Cross Applications	Document & Label Management	PHI & Consent Management	PCM003	Read Access Logging for PHI	Every access to PHI data (patient information, PHI documents, and labels) is recorded through Read Access Logging (RAL). The log captures the user who accessed the data, the timestamp, and the specific resource accessed (document URL). RAL events are written to the SAP Audit Log service with 90-day retention. For regulatory compliance (HIPAA, GDPR), customers must archive RAL events externally using the SAP Audit Log Retrieval service.
Cross Applications	Document & Label Management	PHI & Consent Management	PCM004	Consent management	Customer responsibility to secure consent from users and partners is supported through SAP Data Privacy Integration. The product collects Contact Person personal data (User ID, First Name, Last Name) and provides consent form functionality via hardcoded identifier (DataPrivacyConsentTemplate, cgt, Email). Note: the Consent feature is not available when SAP Advanced Therapy Orchestration apps are accessed through Build Work Zone rather than the native launchpad.
Cross Applications	Document & Label Management	PHI & Consent Management	PCM005	Personal data management	SAP Personal Data Manager service integration enables discovery of where personal data exists, personal data record retrieval (searchable by ID or name+email), export (email or download), and deletion requests. The data subject role is Contact Person. Patient Token deletion is handled separately via SAP Data Retention Manager (DRM), supporting blocking when retention periods are still in effect. After deletion, contact name is replaced with "Deleted User" and emails show "Not available". Requires PDM_CustomerServiceRepresentative role.
Master Data & Configuration	Master Data Management	Organization Management	ORM001	Manage Organizations app	The central app for creating and maintaining organizations – any location in the CGT supply chain including commercial centers, manufacturing plants, distribution centers, treatment centers, and hospitals. Each organization is configured with mandatory elements (address type, organization location types, countries/regions) and optional elements (plants, contact persons, business partner roles, sales areas, purchasing organizations, distribution channels, divisions). Master data cannot be deleted once created – only deactivated. All changes are recorded in the change log. Accurate organization master data is foundational in personalized medicine because every treatment order routes live

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					irreplaceable patient cells through a specific chain; an incorrectly configured location could misdirect a therapy batch to the wrong facility, and the immutable-by-deactivation design ensures a complete audit trail for GxP compliance.
Master Data & Configuration	Master Data Management	Organization Management	ORM002	Location types and categories	Organizations are classified by location categories that define their role in the supply chain: Collection Center (COL), Pickup (PUL), Drop-off (SHL), and Distribution Centre (DSC, added in 2502). Location types determine how the organization participates in flow definition. Node flow sections reference a Location ID from the Manage Organizations app or a Location Rule ID from Configure Location Rules. The location type of an organization constrains which shipment steps it can participate in. This classification is critical because therapy logistics are directional and step-specific: a collection center can only appear at the start of a biospecimen flow, while a distribution center can only serve as a transit node – enforcing these constraints at the master data level prevents misconfigured flows that delay time-sensitive living cell shipments.
Master Data & Configuration	Master Data Management	Organization Management	ORM003	Country-specific address formats	The Manage Organizations and Manage Contact Points apps support country-specific address formats that accommodate locality, neighborhood, and other region-specific address fields. Over 45 countries are supported, including Israel, Oman, Bahrain, Brazil, Japan, United Kingdom, Australia, South Korea, Germany, France, and the United States. These addresses are displayed across all shipment and logistics apps (From/To Location Shipper/Consignee, Manufacturing Locations). Correct address formatting is essential for global CGT programs where biospecimens and finished products cross international borders – courier systems and customs documentation require precise, locally formatted addresses to avoid shipment delays that could compromise cell viability. Introduced in release 2405, enhanced in 2605 with 15 additional countries (Lithuania, Netherlands, Belgium, Croatia, Estonia, Greece, Russia, Uruguay, Latvia, Portugal, Romania, Ireland, Slovakia, Libya, Macao).
Master Data & Configuration	Master Data Management	Organization Management	ORM004	Pickup and dropoff time slots	The Manage Organizations app supports preferred pickup and dropoff time slots for locations within an organization. These time slots are optional and help streamline the courier booking process by communicating preferred collection and delivery windows. The time values are passed to courier systems via the courier booking event and displayed on the courier booking and shipment screens. For cell therapies, adherence to collection and delivery windows is not just a logistics convenience but a viability constraint – living cells degrade outside controlled conditions, so coordinating precise handling times with courier providers reduces the risk of

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					temperature excursions and material loss. Introduce release 2506.
Master Data & Configuration	Master Data Management	Organization Management	ORM005	Organization-Location-Therapy Rules	The Manage Organization-Location-Therapy Rules maps treatment centers to their pickup and drop-off locations per therapy. These rules are used during treatment order processing to determine which locations are involved for a given therapy at a given treatment center. The rules link three dimensions: organization (treatment center), location type, and therapy – ensure that the correct logistics path is followed for each treatment program at each site.
Master Data & Configuration	Master Data Management	Material Management	MAT001	Manage Materials app	The central app for creating and maintaining materials (biospecimen or finished product) used in manufacturing and shipping. Each material is configured with basic information (material type and group, weight information, sales organization and distribution channel), plant data (where material is delivered or processed, batch management, disposition relevance, dosage quantity, maximum number of subunits), and subunit configuration per material+plant combination. Materials cannot be deleted – only deactivated. In autologous CGT, each material record represents a category of patient-derived biological tissue or the resulting drug product, and incorrect material configuration directly impacts downstream processes: batch management settings, COI/COC traceability, disposition relevance gates for release, and the subunit maximum determines how many individually tracked containers (cryobags, vials) are created per patient batch.
Master Data & Configuration	Master Data Management	Material Management	MAT002	Subunit configuration	Each material-plant combination can define subunit properties: Subunit ID Mode (Auto Generate via serial number, or Batch via material batch number), Subunit Type (Cryobag, Bag, or Vial), Subunit Qualifier (Cryobag ID, or Vial ID), and Shelf Life (determines batch expiration date). The maximum number of subunits configured on the material determines how many subunit records are created in processing apps. This configuration drives the granularity of traceability – each subunit is independently tracked through the COI/COC chain. Subunit-level tracking is essential because a single patient's therapy may be split across multiple cryobags or vials, each shipped and stored independently, and regulators require proof that every individual container maintains a chain of identity from collection through infusion.
Master Data & Configuration	Master Data Management	Material Management	MAT003	Inbound material determination	The Manage Inbound Materials app maps biospecimen shipment materials based on Country/Region + Patient Weight + Therapy (or Therapy Category) -> Material determination. Rule-based determination is used during flow verification confirmation when the specific biospecimen material is not explicitly provided. The material determination

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					also be automated via the Business Decisions frame (BS/BKS/FS-FL Material Determination type, controlled by feature flag ft_material_determination). Automating inbound material determination reduces manual effort during order creation – in personalized therapies, the collection kit material often varies by patient weight and geography, and selecting the wrong kit could lead to insufficient cell harvest or wasted collection supplies.
Master Data & Configuration	Master Data Management	Material Management	MAT004	Outbound material determination	The Manage Outbound Materials app maps finished product and biospecimen kit shipment materials by Therapy + Material Group + Material Type + Country/Region + Processing Plant + Patient Weight + Material. This six-attribute mapping enables complex material determination logic for therapies where the finished product varies by manufacturing site, patient characteristics, or regional requirements. Like inbound determination, this can be automated via Business Decisions when the feature flag is enabled. The six-attribute granularity is necessary because CGT finished products are not interchangeable across patients; for example, a finished product manufactured at Plant A for a pediatric patient in Japan may differ in formulation, labeling, packaging from one manufactured at Plant B for an adult in the United States.
Master Data & Configuration	Master Data Management	Material Management	MAT005	Manage Plants app	Dedicated app for creating and maintaining plant records, which represent manufacturing or processing locations with specific capabilities. Plants are linked to organizations and materials, forming the foundation for schedule management and ERP document creation. Plant data determines where materials can be processed and manufacturing orders are created on schedule confirmation. In cell and gene therapy, manufacturing capacity at each plant is limited and scheduling is constrained by slot availability, so accurate plant record data is a prerequisite for the scheduling algorithm to correctly identify which facilities can produce a given therapy. This also supports soft booking expiry duration for scheduling. The app was moved under the Master Data section on the launchpad in release 2409.
Master Data & Configuration	Master Data Management	Therapy Management	THM001	Manage Therapies app	The central app for creating and maintaining therapy definitions, which represent the specific cell or gene therapy programs orchestrated by SAP Advanced Therapy Orchestration. Each therapy links a Therapy Type and Therapy Category (from configuration apps) with the organizations that perform the therapy, the countries/regions where it is applicable, and the lab types used. Therapies are the primary classification dimension that drives role-based access filtering (via therapyId attribute on roles) and flow version determination. Because each cell or gene therapy program has unique manufacturing processes, logistics requirements, and regulatory approvals, the therapy

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					master data record serves as the anchor that connects configuration – from which hospitals can order the to which label templates and flow versions apply.
Master Data & Configuration	Master Data Management	Therapy Management	THM002	Therapy-organization-country linking	Each therapy configuration links three dimensions: organizations (with treatment center location type administer the therapy, which countries/regions the therapy is approved for, and which label types are used. This multi-dimensional linking ensures that treatment orders are only created for valid therapy-organization-country combinations, and that the correct label template and flow versions are selected based on these parameters. This validation is critical for patient safety – creating a treatment order for a therapy at a hospital not certified to administer it, or in a country where it lacks regulatory approval, could result in an unauthorized treatment with serious legal and clinical consequences.
Master Data & Configuration	Master Data Management	Therapy Management	THM003	Collection Identifier	Each therapy+label type combination can be configured with a Collection Identifier that controls donor identification during label generation. Two modes are available: EarliestDIN (Earliest Donation Identification Number, the default) and MultipleDINs (Multiple Donation Identification Numbers). This is relevant for Finished Product, Secondary Packaging, and Finished Product Short Label types where the donor identification factor affects the label content and regulatory compliance. Correct donor identification on labels is a chain of identification requirement – the DIN links the finished drug product to the specific apheresis donation, and an incorrect identifier could result in a patient receiving cells derived from the wrong donation event.
Master Data & Configuration	Master Data Management	Flow Definition Management	FDM001	Manage Flow Definitions app	The central app for creating therapy-specific process blueprints that define all logistics and manufacturing from biospecimen collection to final product delivery. Flow definition can have multiple versions as the therapy process evolves. Flow steps are defined with Step ID, Description, Subsequent ID (creating the sequential chain), and Flow Section Type (Node for locations, Flow for shipments). The app includes tabs for flow milestones, and lead times. Flow definitions are the critical configuration artifact in SAP Advanced Therapy Orchestration because they encode the entire therapy manufacturing and logistics blueprint – every step from cell collection to final product delivery – and a misconfigured flow definition could result in missing manufacturing steps, incorrect shipment routing, incorrect ERP document creation. See Solution Area 2.1 (Therapy Management) for flow version confirmation and preview flow details. The Generate Milestones function can generate both standard and custom milestones (if configured and active) for all relevant entities (shipment and processing activity types) in a flow version. This

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					used to configure lead times between milestones for scheduling
Master Data & Configuration	Master Data Management	Flow Definition Management	FDM002	Manufacturing scenario modeling	Flow definitions support three standard manufacturing scenarios: Simple (biospecimen -> single processing -> finished product -> delivery), Multi-plant (biospecimen -> multiple processing plants with intermediate product shipments between them), and Transit Legs (finished product passes through one or more intermediate centers via FS-IL before final delivery via FS-FL). Any number of processing plant + intermediate product combinations are supported. The flow step structure model the full complexity of CGT manufacturing network. This flexibility is essential because CGT manufacturing networks are highly variable – some therapies require single processing step at one site, while others involve expansion at one plant, viral vector transduction at another, and quality testing at a third, with intermediate product shipments between each stage.
Master Data & Configuration	Master Data Management	Flow Definition Management	FDM003	Co-product parallel flows	Processing activities can yield a main product plus more co-products, each with its own parallel flow branch via the Additional Step ID field (introduced in 2502). Complex scenarios such as the 15-step example in the Guide, a single biospecimen processing step branches into a direct FS-FL delivery for the main product and a separate plant processing chain with transit legs for the co-product. Main product and co-product flows are processed in parallel, with separate shipments, processing activities, and ERP documents for each branch. Co-product support is important for advanced CGT therapies where a single manufacturing run may yield both the primary drug product and a secondary product (e.g., retained sample for potency testing or backup doses), each requiring independent logistics and quality tracking through COI/COC chain.
Master Data & Configuration	Master Data Management	Flow Definition Management	FDM004	ERP document type per shipment	Flow version definitions configure which ERP document type is created per shipment on schedule confirmation. Defaults are: Purchase Order for biospecimen shipment, Sales Order for finished product final leg and biospecimen kit shipments, and Stock Transport Order for intermediate product shipments. For transit leg shipments, the type can be configured as either Stock Transport Order or Sales Order, providing flexibility for different logistics models. Configurable ERP document types per shipment level are necessary because the commercial and legal relationships vary across the supply chain – biospecimen collection is a procurement relationship (Purchase Order), while finished product delivery to the treatment center is a sales transaction (Sales Order), and inter-plant transfers are internal stock movements (Stock Transport Order). Introduced in release 2502.

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Master Data & Configuration	Master Data Management	Transportation & Carrier Management	TCM001	Manage Transportation Lanes app	Defines the logistics lanes that connect locations in supply chain. Each transportation lane specifies the origin and destination locations, the carrier, and the associated service parameters. Lanes are used during shipment processing to determine which logistics route a shipment follows. Lanes can be automatically determined via transportation lane rules or manually selected by using the “Determine Transportation Lane” button in shipment apps. In CGT logistics, transportation lanes encode constraints such as which carriers are qualified to transport living cells under required temperature conditions on a given route, making lane selection a quality-impacting decision rather than a mere logistics preference.
Master Data & Configuration	Master Data Management	Transportation & Carrier Management	TCM002	Manage Transportation Lane Rules app	Configures the rules for automatic transportation lane determination based on material and shipment attributes. Rules match on material type + material group (or transportation group + packaging type) to select the appropriate transportation lane. When multiple rules match, the most specific rule wins. Lane determination is triggered at flow version confirmation for requested materials or at material allocation for allocated materials and the result populates the Logistics tab of the shipment. Automatic lane determination removes a manual step from time-critical workflows – when a treatment or confirmed and a biospecimen shipment is created, the system can immediately identify the correct carrier without operator intervention, saving valuable hours in the process where cell viability degrades with every delay. The app was moved under the Master Data section on the launchpad in release 2409.
Master Data & Configuration	Master Data Management	Transportation & Carrier Management	TCM003	Carrier management	Carriers (logistics providers used for shipment transportation) are managed as master data entities. Each carrier record contains the carrier’s organization details and service capabilities. Carriers are linked to transportation lanes and are passed to courier systems during the courier booking event. The carrier information is displayed in the Logistics tab of all shipment management apps and in the Courier Booking screen. Carrier qualification is particularly important for cell therapy because not all logistics providers are equipped to handle cryogenic or temperature-controlled biological shipments – the carrier master data ensures that only qualified providers are available for selection on transportation lanes carrying living patient material.
Master Data & Configuration	Master Data Management	Transportation & Carrier Management	TCM004	Manage Inbound/Outbound Materials apps	Two dedicated apps manage the material determination rules for biospecimen (inbound) and finished product (outbound) shipments. These apps define the mapping tables used when material is not explicitly provided during order creation or flow version confirmation. The apps were moved under the Master Data section on the launchpad in release 2409. They are being progressively superseded.

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					the Business Decisions framework (BS/BKS/FS-FI Material Determination type) when the ft_material_determination feature flag is enabled. - migration from static master data tables to BRF+ E Decisions gives customers greater flexibility to enc complex material determination logic – for exampl that consider patient-specific attributes or therapy specific packaging requirements that cannot be ex in a simple lookup table.
Master Data & Configuration	Master Data Management	Transportation & Carrier Management	TCM005	Courier instruction configuration	Configure Courier Instructions application enables administrators to define reusable instruction temp with unique identifiers, names, and default values. Instructions can be associated with specific treatr order types, allowing different courier handling requirements for clinical trial versus commercial or Each instruction and its linked order types can be independently activated or deactivated with mand; reason tracking. All configuration changes are capt change logs for audit compliance.
Master Data & Configuration	Business Configuration	Order & Status Configuration	OSG001	Configure Order Types	Defines the treatment order types available in SAP Advanced Therapy Orchestration, each representir distinct therapy orchestration pattern. Order types linked to flow versions, approval workflows, and sta profiles. The app was enhanced in release 2409 wit Follow-On Order Types section that maps which or types can be used as follow-on orders per therapy, activation/deactivation and default marking. In rele 2506, a Manufacturing Order Type field was added select between Process Order (PI) and Production (PP) for each order type. Order type configuration i foundational because it determines the entire down behavior of a treatment order – which flow version: available, which approval steps are required, which profile governs state transitions, and now which EF manufacturing document type is created. Getting t order type wrong means the patient's therapy follo wrong orchestration path. For a given Order Type, a On Order Type can be configured: It can be the sam Type as its parent or a different Order Type
Master Data & Configuration	Business Configuration	Order & Status Configuration	OSG002	Status profile management	The three-tier status architecture requires compre configuration: User Statuses (configurable display mapped to ~90 SAP-delivered system status code; Status Profiles (groups of user statuses with define transitions – initial status and next status mapping status field), and Entity Status Profiles (assign stat profiles to specific entities, object types, and statu; Fourteen status profile configurations are mandate implementation. Object Type IDs enable per-type configuration (e.g., different status flows for BS vs. shipments, or for different material types). Rigorou management is a GxP requirement in cell and gene – regulators expect that patient material cannot ac

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					to the next processing or shipping step until the previous step's status transition has been properly recorded and authorized, creating an enforceable quality gate at each stage of the therapy journey.
Master Data & Configuration	Business Configuration	Order & Status Configuration	OSG003	Material status model	Materials follow a three-tier status hierarchy independent of the entity status model. Disposition statuses (QI, URU/RWR/BLK) gate quality decisions. Inventory : (AVSHP -> ALSHP -> Shipped) track shipping read Consumption statuses (NCNS -> PCNS -> CNS) track processing usage. These three tiers operate in parallel; a material can be simultaneously Unrestricted Use (disposition), Available to Ship (inventory), and Not Consumed (consumption). The transitions are enforced by the system and drive downstream process eligibility. This three-tier model is essential for patient safety in CRO because it prevents a material that has not passed inspection (disposition) from being shipped (inventory) or consumed in manufacturing (consumption) – each acts as an independent gate, and all three must align before the material can proceed to the next step in the therapy workflow.
Master Data & Configuration	Business Configuration	Order & Status Configuration	OSG004	Reason code configuration	Every save operation across configuration, master data, and transactional apps requires a change reason code, defined by a configured reason code. The Configure Reason Codes and Configure Change Reasons apps manage approximately 100 entity/reason type combinations and 50+ change reason entity mappings. If no reason code is configured for a given entity and action, the save operation fails. This mandatory pattern enforces GxP audit trail and traceability from day one. In release 2409, reason code entity codes were restructured under a "Reason Type" organization.
Master Data & Configuration	Business Configuration	Order & Status Configuration	OSG005	COI type and number range configuration	Configure COI Types defines the Chain of Identity type classifications used in the system. Configure Number Ranges manages the number ranges for system-generated identifiers across entities. Number ranges are tenant-specific and cannot be transported between tenants, ensuring that each environment maintains unique identifier sequences. Both apps are essential for the system setup and must be configured before transaction processing begins. COI type classification and number range configuration directly impact chain of identity integrity – the COI ID is the single most critical identifier for personalized medicine, permanently linking a patient's biological material and drug product throughout their entire therapy lifecycle, and any gap or collision in the number range would undermine the traceability chain that regulators require.
Master Data & Configuration	Business Configuration	Field & Profile Configuration	FPC001	Field control profiles	The Configure Field Control Profiles app controls which fields and field groups are visible in the UI per order type, shipment type, or processing activity type. Admins

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					define profiles with sets of visible/hidden fields, then associate these profiles with specific entities via the Configure Profile Groups app. This enables tailored experiences – for example, clinical trial orders can only relevant fields while commercial orders display a different set. Enhanced in release 2404 with international product shipment fields and document upload/view request profiles. Field control is important for CGT operations because different therapy programs, order types, and user roles need to see different information. A clinical trial coordinator managing investigational therapies needs a different view than a commercial operations specialist, and showing irrelevant fields increases the risk of data entry errors in a process. Mistakes can compromise patient safety.
Master Data & Configuration	Business Configuration	Field & Profile Configuration	FPC002	Copy control profiles	The Configure Copy Control Profiles app (introduced in 2409) controls which optional data sections are copied when creating follow-on orders via the clone operation. Three profile types exist: Treatment Order Copy Control Profile (Business Partners, Order Texts, Order Date Documents, Approval Details), Shipment Copy Control Profile (Documents, Logistics, Approval Info, ERP Data), and Processing Activity Copy Control Profile (Documents, ERP Data). Defaults are shown on the UI but can be overridden by users at clone time. Copy control is particularly important for follow-on orders in CGT, where a patient may need a re-treatment or a therapy adjustment – the system must selectively carry forward the relevant context (business partners, logistics setup) while not copying elements that should start fresh (approval status, documents), ensuring the new order inherits the right baseline without stale or inappropriate data.
Master Data & Configuration	Business Configuration	Field & Profile Configuration	FPC003	Process flow field control profiles	Three additional field control profile types (introduced in 2506) manage the display of attributes in the treatment order process flow View Details popup: Processing Process Flow, Shipment Process Flow, and Treatment Order Process Flow profiles. These are configured via the Configure Control Profiles app and associated with entities via the Configure Profile Groups app using a dedicated Process Flow Field Control Profile dropdown on the Treatment Orders, Shipping Profiles, and Processing Activity Profiles tabs. Each entity can have a single flow profile. Process flow field control enables operational teams to configure exactly which attributes appear when users click into step details on the treatment order flow visualization, reducing information overload and focusing attention on the data points most relevant to each therapy program's monitoring needs.
Master Data & Configuration	Business Configuration	Field & Profile Configuration	FPC004	Profile group management	The Configure Profile Groups app serves as the central association point linking profiles to entities. It manages Field Control Profile groups, Copy Control Profile groups (distinguished by a Profile Group Type field introduced

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					2409), and Process Flow Profile associations. Profile groups also link COI/COC certificate templates and traceability report templates to order types. The app connects the configuration layer to the runtime behavior of all transactional apps.
Master Data & Configuration	Business Configuration	Event & Process Configuration	EPC001	Post-Processing Framework events	The Configure Post-Processing Events app links ~7 business contexts to ~50+ configurable events that follow-on processes when business objects change. Event categories include ERP Integration (Create/Update/Cancel SO/PO/STO/Process Order), Workflow Events (approval, cancellation, termination, printing), Process Steps (automatic flow version confirmation, automatic schedule confirmation, transportation lane determination, expiration date calculation), and COI Events (Notify_COI_Event driver, Monitor COI app). Critical insight: automatic flow version confirmation and schedule confirmation are PPF events that must be explicitly configured. Enhanced in 250, show only relevant events per business context in the dropdown. The PPF is the automation backbone of Advanced Therapy Orchestration – without correctly configured PPF events, critical processes like automatic flow version confirmation, schedule confirmation, document creation, and COI event recording would trigger, leaving the therapy orchestration process slow or incomplete. Understanding that these are explicit configuration steps rather than default behaviors is essential for implementation teams.
Master Data & Configuration	Business Configuration	Event & Process Configuration	EPC002	Configure Rules app	The central registration point for all business rules decisions in SAP Advanced Therapy Orchestration. Rules are configured per Entity (Order, Shipment, Process Activity, Material, Sub Unit, Label, Therapy), Scenario (Flow Version, Material Group/Type, ERP Integration Rule Provider (Workflow or Business Rule), and Rule Provider ID (the deployed rule service or decision ID in SAP Build Process Automation). Approximately 60 entity/scenario combinations are available. The Configure Rules app acts as the bridge between SAP Build Process Automation and SAP Advanced Therapy Orchestration runtime.
Master Data & Configuration	Business Configuration	Event & Process Configuration	EPC003	Configure Destinations app	Manages the BTP destination endpoints that SAP Advanced Therapy Orchestration uses to connect to external services. Key rule providers include: DEFAULT_CONNECTIVITY_URL (DDI add-on), Default Patient Viewer, Default Document Uploader, Default Viewer, IDP_OIDC_URL (e-signature), ESIGNATURE_AUTH_URL (overridable e-signature endpoint), DEFAULT_GBT (batch genealogy), and treatment system destinations for ERP and treatment center platform integration. Destinations are tenant-specific and cannot be transported. Destination management is a critical integration control point because SAP Advanced

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					Therapy Orchestration operates as the orchestrator connecting multiple external systems – ERP for manufacturing orders, treatment center platforms order intake, e-signature services for GxP compliance document management for label and report storage batch genealogy for traceability. Each destination is correctly configured per tenant to ensure data flow to the right systems.
Master Data & Configuration	Business Configuration	Event & Process Configuration	EPC004	Custom text type configuration	The Configure Custom Text Types app (introduced in 2502) defines custom text types with a unique ID and a description. Custom text types can be activated and deactivated and are used across the Additional Information tabs on treatment orders and all shipment types. SAP recommends using unique text IDs per business entity to avoid duplication in outbound events. SAP S/4HANA. Access requires CGT_GeneralData_Config_Read_Srv and CGT_GeneralData_Config_Write_Srv roles. Custom text types enable organizations to capture therapy-specific site-specific supplementary information (such as shipping handling instructions, patient-specific notes, or regulatory references) alongside standard order and shipment data, providing flexibility without requiring schema extensions.
Master Data & Configuration	Business Configuration	Event & Process Configuration	EPC005	Processing Activity Types configuration	The Configure Processing Activity Types app (introduced in 2506) allows creation of customized processing activity types alongside the standard types delivered with the system. Processing activity types are used to differentiate processing activities based on the Flow Section Type. A custom type can be selected in the Configure Flow Section app. The Processing Activity Type field is displayed in the Manage Processing Activities app. Custom types can be downloaded via the Download Entity API; standard types cannot. Custom processing activity types enable organizations to model therapy-specific manufacturing steps beyond the standard types – for example, distinguishing between cell expansion, viral vector transduction, cryopreservation, and quality control as discrete processing activity types, each with its own status profile and field control configuration.
Master Data & Configuration	Business Configuration	Event & Process Configuration	EPC006	Feature flag management	Ten feature flags control the enablement of new capabilities in SAP Advanced Therapy Orchestration managed by SAP via support cases. Key flags include ft_enable_order_clone (follow-on orders), ft_support_scheduling (scheduling), ft_material_determination (automated material determination), ft_release_decision (batch release decision history), ft_cloud_events (Cloud Events migration), and ft_satellite_create (multiple PHI data vaults). Feature flags ensure that new features are activated after customer validation in pre-production environments. This controlled activation model is essential for GxP-validated environments where every system change is meticulously controlled.

Line of Business	Solution Area	Solution Capability	Feature Capability ID	Feature	Short Description
					change must be tested, documented, and approved for production use – enabling a feature flag in pre-production allows quality teams to validate the new capability against their SOPs and validation protocols before it touches patient data.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD001	Two authoring approaches	SAP Advanced Therapy Orchestration supports two approaches for business rule authoring via SAP Business Process Automation (BPA): the older Business Rule Authoring (authored in BPA -> Monitoring -> Business Rules and Data Objects and Rule Services) and the newer Business Decisions (authored in BPA -> Lobby -> Project -> Lobby using Data Types and Decisions, deployed to “Production” environment). Both register identically in SAP Advanced Therapy Orchestration’s Configure Rules app. Business Decisions are the recommended approach going forward as they use the newer BPA Lobby experience and offer exclusive capabilities not available as Business Rule Authoring (Label Template Determination and Material Determination). The ability to externalize business rules into configurable rules rather than hardcoding it is particularly valuable in CGT because therapy programs evolve rapidly – new treatment centers come online, countries receive regulatory approval, and manufacturing networks change, and all of these variations can be handled through rule updates rather than code changes.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD002	Flow Version Determination	Filters available flow versions when a treatment order or flow version is being confirmed. Inputs include The Country, and optionally Treatment Center, Order Type, and custom fields. The output is a list of matching flow versions. This rule is critical because it controls which therapy blueprint is applied to a given order – the wrong flow version could result in incorrect manufacturing or logistics paths. For a patient whose cells are already collected, applying the wrong flow version means that a biospecimen could be routed to the wrong manufacturing plant or processed with the wrong steps, potentially rendering an irreplaceable therapy batch unusable.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD003	Material Group & Type Determination	Determines the material classification used for flow version node matching during flow version confirmation. Inputs include Therapy, Country, Treatment Center, Shipment Type, and Custom Fields. The output is the MaterialGroupType used to match flow step definitions to actual materials. This rule bridges the gap between abstract flow definition and the specific materials available at a given plant. Without this determination, the system could not match the generic “biospecimen” or “finished product” placeholders in the flow definition to the actual material numbers that exist in SAP S/4HANA, which prevent ERP document creation and block the manufacturing process.

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Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD004	Lead Time Determination	Calculates milestone lead times per therapy, count treatment center combination. Inputs are TreatmentCenter, Therapy, and Country; outputs a leadTime (days) and bufferTime (days). These lead drive the scheduling algorithm – given a preferred biospecimen pickup date, the system uses lead time to calculate all downstream milestone dates and check manufacturing slot availability. Accurate lead time determination is critical in CGT because manufacturing slots are scarce and patient treatment windows are clinically constrained – overestimating lead times vs capacity, while underestimating them risks scheduling patient's treatment before the drug product can be
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD005	ERP Document Integration rules	Determines the order type, business partner roles, target system ID for ERP document creation during schedule confirmation. Inputs include treatment order shipment, and therapy attributes. Outputs include OrderPartners with orderType, businessPartnerRole, locationType. This rule controls which ERP document (Sales Order, Purchase Order, Stock Transport Order) is created for each shipment and which business partners are included. Enhanced in 2506 with BRF-based order determination for process orders vs. production order
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD006	Target System Determination	Routes outbound events to the correct external system based on OrderType, Therapy, Country, and TreatmentCenter. The output is a SystemId identifying target. This is essential for scenarios where different treatment center platforms serve different therapy programs or where commercial and clinical orders are processed by different external systems. Enhanced in 2409 to support multiple treatment center platform routing.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD007	Material for Intermediate Shipments	Determines the intermediate product material during version confirmation based on Therapy, Country, OrderType, TreatmentCenter, and Plant. The output is a list of candidate materials. This rule is invoked when the flow definition includes intermediate product shipment and the material needs to be determined dynamically rather than being hardcoded in the flow definition. Dynamic intermediate material determination is important for multi-plant CGT manufacturing scenarios where the intermediate product (e.g., partially processed cell line) shipped between facilities) may vary depending on therapy, manufacturing site, or patient-specific factors
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD008	Quantity Type for Shipment Weight	Determines which quantity type to use for gross weight calculation on shipments. Inputs are ShipmentType, MaterialType, and MaterialGroup. The output is the QuantityType (requested quantity or allocated quantity) used to calculate the shipment's gross weight. This is sent to courier systems via the courier booking request and is needed for transportation planning. Accurate

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					shipment weight is essential for CGT logistics because cryogenic shipping containers (cryoshippers) have weight limits, and courier pricing and handling procedures for temperature-controlled biological shipments are weight-dependent.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD009	Post Processing Pre-Checks	Gates whether a PPF event should actually be triggered by evaluating context-specific conditions and returning a Boolean result. This enables customers to add preconditions to automated event triggers – for example, “only trigger the approval workflow for U.S. orders” or “send the exception analysis event for high-priority therapies.” Without a pre-check rule returning true, the event is suppressed. Pre-checks provide essential differentiation for organizations managing multiple therapy programs with different operational requirements – a global pharmaceutical company might need approval workflows only for certain countries or therapy types, and pre-checks enable differentiation without requiring separate system configurations for each program.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD010	Expected Yield Material / ERP Order Type	Determines the yield material and ERP order type for sequential processing activities. Inputs include treatment order attributes and processing node details (Plant, Organization, Processing Activity Type). Outputs are ERPOrderType and MaterialID (0 to many yield materials). Enhanced in 2506 to determine yield material for sequential processing activities (PA followed by PA shipment) and to use BRF rules for ERP order type (process order vs. production order) instead of iFlow-based determination. Yield material determination is critical for multi-step CGT manufacturing where a processing activity may produce different output materials depending on the therapy, manufacturing site, or processing step – for example, a T-cell expansion at Plant A might yield a different intermediate product specification than the same step at Plant B due to different equipment or process parameters.
Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD011	Label Template Determination (Decisions only)	Determines the label template and language for production operations. Inputs are therapyId, labelType, countryId, plantId, and materialId. The output is a list of matched LabelTemplate records. This Business Decision type is exclusive to the newer Decisions approach and can be authored as a Business Rule. It automates the label template selection that would otherwise require manual configuration via the Map Label Template tool. Therapy/Country/Region/Material master data are used for automated label template determination, which is particularly valuable as therapy programs scale globally – each country may require different label languages, regulatory content, or format requirements, and manually maintaining the template-to-therapy-country map becomes error-prone when a single therapy is applied across dozens of markets.

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Master Data & Configuration	Business Rules & Decisions	Business Rules & Decision Management	BRD012	BS/BKS/FS-FL Material Determination (Decisions only)	Automated material determination that replaces the manual Manage Inbound/Outbound Materials master data apps. Inputs include TreatmentOrder, Shipment, MaterialGroup, MaterialType, and CustomTexts. This is a MaterialID. This Business Decision type requires feature flag ft_material_determination to be enabled. It is triggered at flow version confirmation for biospecimen kit, and finished product final leg ship. When not enabled, the existing material determination master data app remains unchanged. This Business Decision represents the strategic evolution from static lookup tables to dynamic, rule-based material determination – enabling organizations to incorporate treatment order context (including custom text fields) into the material selection logic, which is essential for complex therapy programs where the same shipment type may require different materials based on patient-specific or order-specific attributes. Introduced in release 251.
Quality & Compliance	Chain of Identity & Custody	COI/COC ID Management	CID001	COI ID generation rules	COI IDs are generated via configurable rules supporting two patterns: Sequential (start/end sequence number, IDs increment by one) and Random (per-character position types including Fixed, Numeric, Alphanumeric, and Current Year, with exclusion criteria and a maximum identical-characters constraint). Only one default rule can be active at a time, and deactivated rules cannot be reactivated. Configuration is managed via the Configuration COI Generation Rules app, which is tenant-specific and cannot be transported between tenants. The COI ID is the single most critical identifier in autologous cell therapy; it is the permanent, unbreakable link between a patient and their biological material throughout the entire therapy lifecycle. A COI ID collision or generation failure could mean two patients share an identifier, creating a potentially fatal mix-up risk. The constraint that deactivated rules cannot be reactivated ensures that when a numbering scheme is retired, it cannot accidentally be brought back into use, protecting against identifier reuse. An optional Checksum is added to support ISBT 12 standard in 2605 release. The Checksum is applicable for 15-character COI Identifier.
Quality & Compliance	Chain of Identity & Custody	COI/COC ID Management	CID002	15-character COI ID with year code	The COI ID length was extended from 12 to 15 characters in release 2506, with support for a two-digit current year code indicator at any position. When users select a current year at a given position, the system automatically reserves the next position for the second digit of the two-digit value (e.g., selecting current year at position 3 for 2026 places “2” at position 3 and “5” at position 4). The last position cannot be selected as the year code since two digits are needed. The COI ID generation API follows rule configuration to generate identifiers of the configured length. The extension to 15 characters and the year code support address a practical scaling requirement –

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					therapy programs grow and more patients are treated over multiple years, longer identifiers with embedded year codes reduce the risk of exhausting the number range and make it easier to visually identify when a COI ID was generated during audits and investigations.
Quality & Compliance	Chain of Identity & Custody	COI/COC ID Management	CID003	COI/COC ID Creation API	The POST /createCOICOCID API creates a unique COI/COC ID against a Patient Therapy Number (PTN). A request contains a PTN that already has a COI/COC ID. The system returns the existing ID along with the times when it was generated, preventing duplicate ID creation. This API is the primary programmatic interface for treatment center platforms and ordering systems to obtain COI/COC identifiers for new therapy orders by submitting them to SAP Advanced Therapy Orchestration. The idempotent design (returning existing IDs for the same PTNs rather than creating duplicates) is a critical system feature – in a distributed system where treatment center platforms and SAP Advanced Therapy Orchestration communicate asynchronously, network retries or duplicate requests must never result in multiple COI IDs for the same patient therapy, as this would fracture the chain of custody. Enhanced in 2605 to accept optional COI Identifier Generation Rule to choose how the COI Identifier is generated.
Quality & Compliance	Chain of Identity & Custody	COI/COC ID Management	CID004	External COI/COC ID replication	The CoI/CoC ID Replication inbound event (sap/cgt/ext/coicoc/id/replicate) enables replication of externally generated COI/COC ID into SAP Advanced Therapy Orchestration without creating it against an internal generation rule. The event supports replication with or without reference documents (Patient Therapy Number, Therapy, etc.) and validates uniqueness – duplicate IDs are rejected. An "Is COI ID External" flag in the Monitor COI app and Manage Orders distinguishes externally generated IDs from CGTO-generated ones. External ID replication is essential for organizations that have existing COI ID generation systems (such as 128-based identifiers used in blood banking) and need to integrate with SAP Advanced Therapy Orchestration to honor those identifiers rather than generating new ones – maintaining a single consistent identifier across all systems in the therapy supply chain is a regulatory expectation. Introduced in release 2409. Enhanced in 2605 to accept optional Identifier Generation Rule and Checksum to support the 128 standard and allow customer to validate the COI Identifier against specific COI Identifier pattern.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV001	Comprehensive event model	Every state change across the treatment order lifecycle generates a COI or COC event, organized by business object group: Treatment Order (COI Creation, Order Creation), Biospecimen Shipment (Shipment, Collection, Material Batch/Subunit Receipt, Disposition, Consumption, Usage Decision), Processing/Manufacturing (Creation, Start, End, Yield Material Batch/Subunit

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					Disposition), Intermediate Product Shipment (Ship Batch Allocation, PGI/PGR, Disposition, Consumpt and Finished Product (Transit/Final Leg Shipment, Allocation, PGI/PGR, Disposition, Receipt at Treatn Center). Events are NOT triggered for initial status from blank to Quality Inspection – only meaningful transitions are tracked. This comprehensive event the foundation of regulatory compliance in CGT – regulators such as the FDA and EMA require compl traceability of every action taken on patient materi the moment cells are collected to the moment the product is administered. The event model ensures meaningful state change goes unrecorded, creating audit trail that proves a specific patient's cells were handled correctly at every step.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV002	COI event profile configuration	The Configure COI Event Profiles app controls which events and attributes are visible in the Monitor Chain of Identity app and traceability reports. Each profile lists internal event types (enabled by default) and allows enabling/disabling individual event types and their attributes. Profiles are linked to order types via the Configure Profile Groups app. When a new event type is added (such as the State attribute introduced in 2506), customers must create a new COI event profile to select the monitor. Profile-based visibility control is important because different therapy programs or organizations may require different levels of COI event detail – a manufacturing quality manager may need to see all processing-related events and attributes, while a logistics coordinator only needs shipment events. Event profiles also govern what appears in traceability reports, ensuring that generated compliance documentation matches the organization's reporting requirements.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV003	COC event profile configuration	The Configure COC Event Profiles app provides the profile-based control for the Monitor Chain of Custody app. COC profiles are maintained separately from COI profiles, enabling independent control over which chain events and attributes are visible. The same profile-driven approach applies to the traceability report – and attributes disabled in the COC profile are excluded from the COC section of the report. Label changes renamed “Event” column to “Event Type” and “Parent Event” to “Parent Event Type” in both COI and COC apps. Maintaining separate COI and COC profiles is architecturally sound because identity tracking (who the patient) and custody tracking (who physically held material) serve different regulatory purposes and audiences – a quality assessor investigating a COI issue needs to see different event details than a logistics manager reconstructing the physical chain of custody.
Quality & Compliance	Chain of Identity &	COI/COC Event	CEV004	External event type registration	The Configure External Event Types app (renamed “Configure External Events” in 2506) enables registration

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	Custody	Management			of events from non-SAP systems in the COI/COC chain. Each external event type can be associated with a Parent Event Type for hierarchical organization. Enhanced inbound validation ensures that the event type and business object in incoming payloads align with the configured settings. Access requires CGT_CoiCocAdmin_App, CGT_CoiCocAdmin_Read and CGT_CoiCocAdmin_Write_Srv roles. External event type registration is critical for organizations with multiple vendor therapy supply chains – manufacturing steps performed by CDMOs, quality testing by third-party or cryogenic storage by specialized providers all generate traceability events outside SAP Advanced Therapy Orchestration that must be captured in the unified COI/COC chain to maintain an unbroken audit trail. Introduced in release 2502.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV005	External attribute management	The Configure External Attributes app manages attributes not in SAP Advanced Therapy Orchestration's predefined attribute set, enabling customers to define traceability data points specific to their non-SAP systems. All custom attributes accept values in string format only. The app includes a tab displaying pre-delivered SAP attribute reference. External attributes are associated with event types and are validated during inbound event processing. Custom external attributes enable organizations to capture traceability data points that are specific to their non-SAP systems and not covered by SAP Advanced Therapy Orchestration's standard attributes. For example, a CDMO's internal lot number, a cryogenic provider's tank and canister position, or a courier's tracked temperature readings. Introduced in release 2502.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV006	External events in event profiles	Starting with release 2506, external event types can be added to COI and COC event profiles via a new External Events section. Users add one external event type at a time, and Parent Event Type and Group details are populated as read-only fields. External event types can only be added to a profile unless their parent event type is already registered within the same profile. Users can enable or disable external event types to control their visibility in the Monitor apps and traceability reports. Navigation from the event profile to the Configure External Event Types app is supported. Including external event types in event profiles closes the loop on external traceability integration – without this capability, external events could not be recorded in SAP Advanced Therapy Orchestration, making them invisible in the Monitor apps and traceability reports, creating gaps in the compliance documentation that regulators review during inspections.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV007	Col/CoC Creation inbound event	The Col/CoC Creation event (sap/cgt/ext/coicoc/event/create) enables external systems to create COI or COC events in SAP Advanced Therapy Orchestration. The event payload was enhanced in release 2506.

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					2506 with new fields (materialDisplayId, subUnitId, batchNumber, rootBusinessObjectId, additionalInfo). Additional validations including checks against previous external event type configurations and parent event alignment. This is the primary interface for integrating non-SAP traceability events into the SAP Advanced Therapy Orchestration compliance chain. The enhanced validations in 2506 are important because invalid or misaligned external events could corrupt the COI/COC chain – for example, an external event referencing a non-existent event type or mismatched business object could create misleading traceability records that undermine regulatory integrity of the entire chain.
Quality & Compliance	Chain of Identity & Custody	COI/COC Event Management	CEV008	New event attributes (2511)	Four new attributes were added to COI and COC event profiles in release 2511: Source System ID (SourceSystemID, common across all events – identifies the triggering system), Shipment Material Expiration Date (ShipmentMaterial.expiryDate, for collection material batch and allocation events), Treatment Donor ID (TreatmentOrder.donorId, for Order Creation COI events) and Shipment Material Dosage Quantity (ShipmentMaterial.dosageQuantity, for FS-FL allocation events). Source System Identifier is included in previously delivered traceability reports; other fields must be added to COI/COC Certificate Templates by customers. The Source System ID attribute is particularly significant in multi-system CGT environments – when COI/COC events originate from different systems (SAP Advanced Therapy Orchestration, SAP S/4HANA, external CDMOs), knowing which system triggered each event is essential for root cause analysis during deviation investigations and demonstrating system integration integrity to regulators.

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Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO001	Monitor Chain of Identity (COI) – Graphical View	The graphical view displays COI events grouped as in a hierarchical flow: Treatment Order -> Biospecimen Processing -> Intermediate -> Finished Product. Events are displayed as child nodes under their parent event type (if configured) or as independent nodes (parent). Only the latest event per event type is shown in the graphical view. Batch genealogy information can be retrieved in real-time from the Global Batch Trace (GBT) system via SAP Batch Release Hub, displayed alongside production end events. Multiple Treatment Order numbers are shown for follow-on orders to represent both the inactive and active orders. The graphical view is the primary compliance monitoring interface for quality and regulatory personnel – it provides an at-a-glance visual confirmation that a patient's therapy has progressed through every required step, from biospecimen collection through manufacturing to final product delivery, with no gaps in the identity chain. Real-time batch genealogy retrieval from the GBT system enriches this view with manufacturing lineage data that would otherwise require switching to a separate system.
Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO002	Monitor Chain of Identity (COI) – Tabular View	The tabular view displays all events without aggregation (both create and update actions visible), providing a complete chronological record. An Event Attributes view displays event-specific attributes in JSON format, accessible via the View Event Attributes button. Events can be exported to Excel via the Export function. Ten filters were added in release 2506: Event Type, Business Object ID, Business Object ID (enabled by default), plus Batch Number, Batch Created By/On, Material, Root Object ID, Subunit, and Treatment Order (available via Adapt Filters). The tabular view complements the graphical view by providing a complete, unfiltered chronological record needed for detailed investigations – when a quality event or deviation occurs, investigators need to see every action (including updates and corrections) in sequence, not just the state per event type. The Excel export capability supports audit documentation workflows where traceability data must be extracted and attached to deviation reports for regulatory submissions.
Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO003	Monitor Chain of Custody (COC)	The Monitor Chain of Custody app tracks the physical custody chain – who had custody of what material, when, and where – across the supply chain. It mirrors the monitor's dual-view architecture (graphical and tabular) with the same event profile-driven visibility control, external event integration, and export capabilities. COC events are managed through separate COC event profiles. The tabular view provides the same filter and Event Attributes capabilities as the COI monitor. Enhancement 2506 added external events in graphical and tabular views. The separation of COC from COI monitoring reflects a fundamental regulatory distinction: COI proves that

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					right patient's cells were used to make the right pa drug product (identity), while COC proves that the was handled by authorized personnel under contro conditions at every handoff point (custody). Both c must be independently verifiable during regulatory inspections.
Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO004	Usage Decision tracking in COI	A Usage Decision (UD) node in the Monitor COI app linked to batch nodes for biospecimen batches, pro activity yield batches, intermediate product batches, transit leg batches. The node captures the usage d code and text from batch release events triggered by PPF mechanism. Key displayed fields include COI ID, Material ID, Batch Number, UD Code, UD Text, Deci On, and Decision Set By. Multiple usage decision n can exist per batch when multiple release decision: made. Embedding usage decisions directly in the C monitor creates a unified compliance view – quality personnel can see not just that a batch moved thro supply chain, but also the release decisions that authorized each transition, without needing to cross reference a separate batch release system. This is particularly valuable for multi-level release scenarios where different quality authorities may need to app the same batch at different stages. Introduced in re 2502.
Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO005	Follow-on order event handling	When a follow-on treatment order is created, COI a events associated with cloned business objects are to the follow-on order. A new State attribute (active/inactive) on root-level events indicates whe business object belongs to the active or inactive or CLONE action event is recorded at root level for all business objects. Copied events retain their original timestamps and are displayed in the graphical view follow-on order but are excluded from the tabular v traceability report. Customers must create new eve profiles to see the State attribute. Follow-on order handling is essential for patient retreatment scenarios when a patient needs a second round of therapy, th treatment order must carry forward the complete traceability history from the original order so that regulators and quality teams can see the full patier journey across both treatment cycles, while the active/inactive state ensures that the current orde events are clearly distinguished from historical one
Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO006	COI ID validations in SAP S/4HANA	Robust COI ID validations during goods movement S/4HANA ensure the therapy is administered to th correct patient. Validation points include: biospecir receipt at plant, transfer from storage to production process/production order release, biospecimen ma consumption, batch picking for delivery, and batch shipment. The batch characteristic name used for ID is configurable, and validation messages can be

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					overridden from error to warning or info severity. Validations are non-blocking when the COI ID is not in documents. These S/4HANA-embedded validations are a critical patient safety feature – they enforce COI verification at the physical manufacturing level where warehouse operators and production staff handle a patient material. Without these validations, a warehouse operator could accidentally pick the wrong batch for production order or ship the wrong patient's drug product with potentially fatal consequences in autologous therapies. Introduced in release 2511.
Quality & Compliance	Chain of Identity & Custody	COI/COC Monitoring	CMO007	MIGO COI enhancements	The MIGO transaction in SAP S/4HANA has been enhanced with COI reference fields for Movement Type 343 (Disposition) and Movement Type 322 (Reversal). Both events are transmitted to SAP Advanced Therapy Orchestration for traceability. Additionally, material document display in MIGO shows read-only CGTO (COI ID, shipment ID, subunit info) for goods receipt, goods issues, and their reversals related to CGTO creation. These enhancements embed COI traceability directly into the warehouse operator's daily workflow, eliminating the need to switch between SAP S/4HANA and SAP Advanced Therapy Orchestration to verify identity chain information during routine goods movements. For disposition creation (Movement Type 343) and their reversals (Movement Type 322), capturing the COI reference ensures that quality decisions made at the warehouse level are immediately reflected in the end-to-end traceability chain. Introduced in release 2511.
Quality & Compliance	Chain of Identity & Custody	Traceability & Compliance Reporting	TCR001	Traceability report generation	The Generate Traceability Report button on the Documents tab of the Manage Orders app produces a report of all COI and COC events for a treatment or order. Reports use configurable templates (.xdp format) with SAP logo and watermark. Multiple report versions can be generated for the same order, and reports can be saved in the app and downloaded as PDF. Requires SAP BTP Form Service and SAP BTP Object Store (AWS S3 compatible). Only active orders are supported. The traceability report is the definitive compliance document for a patient's treatment – it serves as the auditable proof that the chain of identity and custody was maintained from cell collection through drug product delivery. Regulatory inspectors from FDA, EMA, and national health authorities may request this report for any treated patient, making on-demand report generation with template-based formatting an operational necessity for commercial-scale therapy programs. Introduced in release 2502.
Quality & Compliance	Chain of Identity & Custody	Traceability & Compliance Reporting	TCR002	Profile-driven report content	Starting with release 2506, the traceability report content is controlled by COI and COC event profile configuration. Events disabled in profiles are excluded from the report, and attributes disabled in profiles are also excluded. Display/print combinations are supported: all COI and COC events, COI and COC events with attributes, and COI and COC events with attributes and subunits. Introduced in release 2506.

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					with one COC root event, all COC events with one COI event, all COI and COC events, or selected COI and COC events only. If no event profiles are configured for the event type, all events and attributes are shown for backward compatibility. Common attributes (Event Name, Description, Date/Time, Performed By, Material, Batch, etc.) cannot be controlled via profiles. Profile-driven content ensures that traceability reports contain exactly the information needed for their intended audience. An external regulatory submission report for a regulatory submission may need the complete COI and COC event chain, while an internal quality report might only require COI events with selected attributes. Four display/print combinations provide flexibility without requiring custom report templates for each use case.
Quality & Compliance	Chain of Identity & Custody	Traceability & Compliance Reporting	TCR003	COI certificate generation	COI certificates contain patient, biospecimen, and product information and are generated for Finished Product Leg (FS-FL) shipments at the shipment or batch level. Generation requires a configured certificate template mapped to the order type (via Configure Profile Group) and a valid e-signature from the generating user, and the SAP Document Integration add-on for document storage and rendering. The CGT_CoiCoc_Document_Action role authorizes certificate generation from the Monitor Orders app. COI certificates are the formal compliance documents that accompany finished product shipments to treatment centers, providing the treating physician with verifiable proof that the drug product was manufactured from the correct patient's cells and maintained chain of identity throughout. The e-signature requirement ensures that certificate generation is an accountable, GxP-compliant action with a clear audit trail of who certifies the identity chain.
Quality & Compliance	Chain of Identity & Custody	Traceability & Compliance Reporting	TCR004	Batch genealogy	In the Monitor COI app, production end events can retrieve batch genealogy information in real-time from the Global Batch Trace (GBT) system via SAP Batch Release Hub for Life Sciences. This provides upstream and downstream traceability of how batches relate to each other across manufacturing steps. The feature requires an HTTP destination configured for the GBT system (using the DEFAULT_GBT rule provider) and the Batch Genealogy Information attribute enabled in the COI Event Profile for the Production End event. Batch genealogy extends traceability into the manufacturing dimension – while events track the logistics and status chain, batch genealogy reveals the material composition chain: which input batches (biospecimen, reagents, vectors) were consumed to produce which output batches, and how those output batches relate to downstream processing steps. This cross-system integration between SAP Advanced Therapy Orchestration and SAP Batch Release Hub provides the complete traceability picture that regulators expect for cell and gene therapy products.

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Quality & Compliance	Chain of Identity & Custody	Traceability & Compliance Reporting	TCR005	Subunit label standardization	Release 2511 standardized subunit label terminology across the platform: “Shipment Subunit ID” was re to “Shipment Line Item Number” and “Shipment R Subunit ID” was renamed to “Shipment Subunit ID” renaming aligns the display labels with the actual c semantics – the Line Item Number identifies the p within the shipment while the Subunit ID identifies physical subunit. These changes affect COI/COC e displays, traceability reports, and all apps where su identifiers are shown. Consistent, semantically acc terminology across the platform is essential for compliance documentation – when a regulatory in reviews a traceability report, the field labels must unambiguously identify what each value represent previous naming created confusion between the sh position number (a logistics concept) and the phys subunit identifier (a traceability concept), which th standardization resolves.
Manufacturing	Processing Management	Processing Activity Management	PAM001	Manage Processing Activities app	The central app for viewing and managing all past, planned, and future manufacturing processes with treatment order. Each processing activity represen manufacturing step defined in the flow version – su isolation, transduction, expansion, formulation, or cryopreservation – with general data, locations, ma batches, subunits, documents, and status informat app displays the Processing Activity Type field to differentiate activities, and the State field (Active/I to support follow-on order handling. In personalize and gene therapy manufacturing, where every batc patient-specific and each step operates under stric viability windows, this consolidated view is essentia Treatment Order Managers to monitor manufactur progress and identify delays before they jeopardize viability. All shipment processing apps were conver from scroll-based to tabbed layout in release 2409 performance.
Manufacturing	Processing Management	Processing Activity Management	PAM002	Create Subsequent processing activity	The Create Subsequent button in the Manage Order creates an additional processing activity at the san step, copying data from a selected parent entity. Co data includes general data (Processing Activity, Typ Organization and Plant, flow version details (flow v step, type), PA locations (organization, plant, addre contact persons), and yield materials (new line iter Material ID, Vendor Material, Dosage Quantity, UoM materials copied only for CMO type). Batch IDs and subunits are regenerated per master data settings. capability supports CGT manufacturing scenarios : batch splitting, re-expansion of cells that did not m potency thresholds, or transferring a manufacturin to a different CMO – situations where the original f definition did not anticipate the need for an additio activity at the same step. Introduced in release 240

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Manufacturing	Processing Management	Processing Activity Management	PAM003	Cancel Processing Activity	Processing activities can be cancelled via the existing Status Transition event, which triggers an outbound message to cancel the associated process or production order in SAP S/4HANA. On cancellation, an automatic exception is generated with source "Internally Generated", Priority "High", Type "Error", and Exception Type "Functional Error". Cancelled processing activities are excluded from material consumption when follow-on orders are created. This is critical in clinical workflows where a manufacturing step may need to be aborted due to cell contamination, viability failure, or a patient's clinical status change – the automatic exception and ERP order cancellation ensure that all downstream systems are immediately notified and no further resources are consumed for a terminated process. Introduced in release 2502.
Manufacturing	Processing Management	Processing Activity Management	PAM004	Processing Activity Create/Update events	Two inbound events enable external system integration with Processing Activity Create (sap/cgt/ext/shipment/processing_node/create) and Processing Activity Update (sap/cgt/ext/shipment/processing_node/update). The create event creates a subsequent processing activity based on the reference of an existing activity and includes an Auto Confirm and Schedule boolean attribute, and the update event updates an existing processing activity by ID. These events enable treatment center platforms and LIMS to manage processing activities programmatically, which is essential in CGT manufacturing where cell processing decisions are often made in real time based on assay results – for example, triggering an additional expansion cycle when potency testing indicates insufficient cell yield. Introduced in release 2502.
Manufacturing	Processing Management	Processing Activity Management	PAM005	Active/Inactive state management	The processing activity header displays a State field with values Active or Inactive. All entities are created with Active by default. When a follow-on treatment order is created, the parent order's processing activities (along with all other entities) are marked Inactive. No changes are allowed on Inactive entities, and all inbound events are automatically routed to the active order's entities. The state model supports the exception handling workflow for therapy reclassification and flow version changes – for example, when a patient's condition changes mid-manufacturing and the therapy must be switched from commercial to a clinical protocol, requiring a new treatment order with a different flow version while preserving the full history of the original order. Introduced in release 2409.
Manufacturing	Processing Management	Processing Activity Management	PAM006	Processing Activity Information API	The Processing Activity View API (GET /ProcessingNodeView) retrieves all processing activities with COI ID, Treatment Plant ID, Plant Location, and Processing Activity Status. Individual activities can be retrieved by ID. The API was significantly enhanced in release 2502 with new \$expand options (shipmentMaterial, document) and 15+ new fields.

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					including flowVersion, processingTypeID, slot confi fields, and contact name fields, enabling external s to access comprehensive processing activity data. is a key integration point for treatment center platf that need real-time manufacturing status to comm therapy timelines to patients and treating physical Standard & custom milestones are also part of the
Manufacturing	Processing Management	Processing Activity Management	PAM007	Receive Milestone updates	Introduced support for receiving and updating both standard and custom milestones in CGTO from external systems via the new sap/cgt/ext/shipment/milestone/update event.
Manufacturing	Processing Management	Manufacturing Order Management	MOM002	Manufacturing Order lifecycle events	The full manufacturing order lifecycle is managed through bidirectional events: Manufacturing Order Creation (outbound to SAP S/4HANA on schedule confirmation), Manufacturing Order Update (outbound on data change), Manufacturing Order Cancellation (outbound on production cancellation), and Manufacturing Order Creation Request (inbound from SAP S/4HANA with status). All events support both process orders and production orders. The creation event includes orderType (from BRF rule) and orderCategory (from treatment order type) fields. Line ID, Study ID, and Protocol ID are included in creation and update payloads (2506), ensuring that the manufacturing floor always has full traceability back to the patient clinical protocol – a regulatory requirement for autologous therapies where each batch is manufactured for a specific patient.
Manufacturing	Processing Management	Manufacturing Order Management	MOM003	CGTO Information tab in SAP S/4HANA	The CGTO Information tab has been enabled in production order screens (CO01/CO02/CO03) and process order screens, displaying Donor ID, Study ID, and Protocol ID as read-only fields received from SAP Advanced Therapy Orchestration. For process orders with co-products, users can view all materials and subunit details in the CGTO Information section, referenced by line item number. This tab bridges the gap between the orchestration layer (Advanced Therapy Orchestration) and the manufacturing execution layer (SAP S/4HANA), giving production operators direct visibility into which patient's therapies are manufacturing – critical for preventing mix-ups at facilities that process multiple autologous therapies simultaneously. The tab was introduced for production orders in release 2506, while process order and other document tabs were enhanced progressively from release 2404.
Manufacturing	Processing Management	Manufacturing Order Management	MOM004	Screen renaming for generic terminology	To accommodate both process orders and production orders, several UI labels were generalized in release 2506: "Process Order Batches" -> "Processing Activity Batches", "Process Order Yield" -> "Processing Activity Yield", "Process Order Components" -> "Processing Activity Components", "Process Order" label -> "Manufacturing Order" (in Report Processing Start/Completion and

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					Manage Product Disposition), and “Manage Process Batches” app -> “Manage Processing Activity Batch new “Manufacturing Order Type” column and “Order Number” column were added. This terminology generalization ensures that the UI is accurate regardless which ERP order type underpins the manufacturing reducing confusion for users working across both production-order-based and production-order-based therapies
Manufacturing	Processing Management	Manufacturing Order Management	MOM005	Configure Processing Activity Types	A new configuration app introduced in release 250H creation of customized processing activity types along with the standard types delivered with the system. Processing activity types are used to differentiate processing activities based on the Flow Section Type in the Configure Flow Section app, enabling organizations to model the specific manufacturing steps in their CGT workflows – such as distinguishing between autologous cell isolation, viral vector transduction, and CAR-T expansion as separate types with distinct behaviors. The Processing Activity field is displayed in the Manage Processing Activities Custom types can be downloaded via the Download API, while standard types are predelivered and can be downloaded. Access requires CGT_GeneralData_Config_Read_Srv and CGT_GeneralData_Config_Write_Srv roles.
Manufacturing	Processing Management	Manufacturing Milestone & Status Management	MMS001	Production milestones from SAP S/4HANA	Manufacturing start, end, and termination milestones captured in SAP S/4HANA via transactions COR6 (time confirmation), CORZ (collective confirmation (capacity overview confirmation), and CORS (cancel and sent to SAP Advanced Therapy Orchestration via Production Milestone inbound event. Upon receipt, processing activity is updated with the actual processing start date, actual processing end date, or termination status. Both process orders and production orders are supported (production order support added in 250H). These milestones are the primary mechanism for synchronizing actual manufacturing progress from shop floor back to the orchestration layer, enabling Treatment Order Managers to track whether production on schedule relative to the patient’s treatment window downstream delivery deadlines. Milestone events are triggered as outbound events from SAP S/4HANA to SAP Advanced Therapy Orchestration.
Manufacturing	Processing Management	Manufacturing Milestone & Status Management	MMS002	Processing activity start and completion reporting	The Report Processing Start app enables users to initiate the actual start of a manufacturing process, record component batch information for intermediate products and biospecimen materials, and view component batch sources (which shipment the component came from). The Report Processing Completion and Yield app captures actual processing end, yield batch information, and enables editing of the expiration date for yield materials. Both require e-signature for GxP-relevant operations, establishing a signed audit trail of who initiated and

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					completed each manufacturing step – critical for batch record integrity where each patient-specific batch is subject to regulatory inspection. Component material support was enhanced in release 2404, and co-product component consumption from previous PAs was addressed in 2502.
Manufacturing	Processing Management	Manufacturing Milestone & Status Management	MMS003	Processing activity status lifecycle	Processing activities follow a defined status lifecycle: Processing Scheduled (ST030) -> Processing Started (ST029) -> Processing Completed (ST034) or Processing Terminated (ST031), eventually reaching Closed (ST035). Status transitions are governed by entity status profiles and require configured reason codes. Processing termination triggers an automatic high-priority exception and cancellation of the associated ERP manufacturing order – this is a critical safety mechanism in CGT, as a terminated process (e.g., contamination detected or equipment expansion) must immediately halt all downstream activities and alert the care team that the patient may need to restart the therapy journey. The Formulation Creation outbound event (sap/cgt/shipment/processing_node/actual_date) is triggered when the actual processing start date is updated.
Manufacturing	Processing Management	Manufacturing Milestone & Status Management	MMS004	Consumption and yield events from SAP S/4HANA	During the processing of a process or production order in SAP S/4HANA, consumption and yield events are created and sent to SAP Advanced Therapy Orchestration via the Production Goods Movement inbound event. On the Advanced Therapy Orchestration side, these events trigger the corresponding processing activity data including component consumption quantities and yield batch information, maintaining a real-time record of what materials were consumed and what was produced at each manufacturing step – essential for the chain of custody and for calculating whether sufficient yield exists to proceed with the patient's therapy. The event was enhanced in release 2506 with an expiryDate field and materialItem, enabling batch expiry date updates from SAP S/4HANA when material is produced. If the expiry date is received via this event, the CGTO-side logic for calculating expiry from the Processing Date is not triggered.
Manufacturing	Processing Management	Batch & Disposition Management	BDM001	Manage Processing Activity Batches app	Formerly "Manage Process Order Batches" (renamed in 2506), this app provides a comprehensive view of a batch's information for a processing activity. Users can view batch details, subunit information, and disposition. The app also displays the Manufacturing Order Type (Process Order PI or Production Order PP) and Order Number columns. Batch IDs can be auto-generated based on master data configuration or assigned manually. In manufacturing, every batch is patient-specific, making this app the central point for verifying that the correct material is used.

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					and subunits are associated with the right patient ID, proceeding to disposition and shipment.
Manufacturing	Processing Management	Batch & Disposition Management	BDM002	Manage Product Disposition app	Formerly “Manage Finished Product Disposition” (renamed in 2404), this app enables users to manage the disposition status of yield material batches from processing activity. The three-tier material status model applies: Disposition (QI -> URU/RWR/BLK), Inventory (AVSHP -> ALSH -> Shipped), and Consumption (NCNS -> PCNS -> Consumed). Disposition changes trigger outbound events to notify external systems and update the COI/COC chain. Electronic signature is required for disposition changes as a controlled relevant operation. The three-tier model is specifically designed for CGT, where a single patient's therapy progresses through quality release (disposition), becomes available for logistics (inventory), and is ultimately administered to the patient (consumption) – each transition requiring controlled authorization and full traceability.
Manufacturing	Processing Management	Batch & Disposition Management	BDM003	Batch disposition and release decision from SAP S/4HANA	Batch disposition information from quality inspection SAP S/4HANA is sent to SAP Advanced Therapy Orchestration via the Finished Product Shipment Event Disposition inbound event, updating the processing activity with the payload data. Release decision history is from SAP Batch Release Hub for Life Sciences capabilities, multi-level release decisions and displays them on Release Decision History tab, providing a consolidated view of quality verdicts from QP (Qualified Person) stability testing, and market-specific requirements which must pass before a patient-specific therapy can ship. The Release Decision Creation event was enhanced in 2502 with usageDecisionCode and usageDecisionDescription fields. Both process orders and production orders are supported (2506).
Manufacturing	Processing Management	Batch & Disposition Management	BDM004	Batch expiry date management	Batch expiry dates can be managed through multiple mechanisms: manual editing via the Report Processing Completion and Yield app (2404), automatic calculation from the Processing Date using shelf life from the master, or receipt from SAP S/4HANA during Post Receipt for process or production orders (2506). A expiry tracking is especially critical in CGT because therapy products have extremely short shelf lives – days for fresh products or months for cryopreserved products – and administering an expired product is a safety risk and a regulatory violation. The SAP S/4HANA provided expiry date is calculated by adding the production date and the shelf life from the material when the PGR document is created. When received from SAP S/4HANA, the CGTO-side calculation logic is suppressed. The expiryDate field was added to outbound events (Batch Status Update, FP Shipment Status) to propagate expiry information to external systems.

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Manufacturing	Processing Management	Batch & Disposition Management	BDM005	Partial consumption between processing activities	Release 2511 introduced partial and full consumption handling units and subunits between sequential processing activities for serialized materials. This scenario where a processing activity consumes or releases the output from a previous activity, with the remainder available for other activities or shipments – a common pattern in CGT where only a portion of expanded capacity can be formulated for the first dose while the rest is cryopreserved for potential re-dosing. The HU-to-SHU sequential mapping from SAP S/4HANA EWM ensures correct tracking of which specific handling units are consumed at each manufacturing step. Non-consumption is also supported for handling units that are returned but not used.
Manufacturing	Processing Management	Batch & Disposition Management	BDM006	Batch allocation to shipments	Manufactured batches are allocated to shipments. Allocate Batches to Finished Product Shipments and Allocate Batches to Intermediate Product Shipments. Allocation updates the inventory status from Available for Ship (AVSHP) to Allocated for Shipping (ALSHP). Deallocation reverses this status. This allocation step bridges manufacturing and logistics – it formally connects a released, patient-specific batch to a specific ship ensuring the correct therapy product is paired with correct delivery to the correct treatment site. New outbound events for FP Batch Details After Allocation and After Deallocate (introduced in 2404) notify external systems when batches are allocated or removed. For products with parallel flows, the flow step identifies previous processing activity's batches are available for allocation (2502). Validation checks on status and for Allocation Events for FSIL & FSFL shipments are ignored, though are now recorded as warning messages in the Monitor Business Events log, rather than block event processing.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM001	Five shipment types	SAP Advanced Therapy Orchestration supports five shipment types, each mapping to a specific ERP document on schedule confirmation: Biospecimen Shipment (BKS Purchase Order), Biospecimen Kit Shipment (BKS Purchase Order, introduced in 2409), Intermediate Product Shipment (IS -> Stock Transport Order, introduced in 2404), Finished Product Transit Leg (FS-IL -> Stock Transport Order, introduced in 2409), and Finished Product Final Leg (FS-FL -> Sales Order). Each type has dedicated management apps, status profiles, authorization roles, and event interfaces. Together they model the complete CGT logistics chain from patient material collection through multi-hop manufacturing to final product delivery.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM002	Shipment status lifecycle	All shipment types follow a consistent status lifecycle governed by entity status profiles: Requested (ST000) -> Scheduled (ST044) -> Shipped (ST080) -> Delivered (ST009) or Partially Delivered (ST010). Additional

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					track cancellation and approval workflows. COI Confirmation Status transitions from Pending (ST040) to Confirmed (ST041) or Rejected (ST043) on shipment receipt based on COI checks. Status transitions require configured reason codes and, for GxP-relevant changes, signatures. A consistent lifecycle across all five shipment types ensures that logistics coordinators apply the operational discipline to every leg of the therapy journey, preventing process gaps that could compromise cell viability or chain of identity.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM003	Shipment creation via events	Shipments are created through inbound events from external systems (treatment center platforms, order management systems). Biospecimen Shipment Create and Finished Product Shipment Create events carry comprehensive payloads including locations, contacts, materials, business partners, courier information, custom texts, and pickup dates. Events support optional fields that fall back to default version logic when not provided (e.g., material ID, pickup location), allowing treatment centers to send minimal payloads while CGTO fills in the details from the core therapy flow – reducing integration burden for hospitals that may have limited IT capabilities. The Auto Confirmation Schedule flag (2502) enables automatic schedule confirmation when the shipment is created.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM004	Subsequent shipment auto-creation	When creating a Finished Product Shipment Final Leg (FS-FL) via UI or create event, the system automatically creates corresponding Finished Product Shipment Leg (FS-IL) shipments if the flow version is already confirmed and the flow definition includes transit legs. This automation is essential for multi-hop CGT logistics; finished therapy products must transit through regional distribution hubs before reaching the treatment center. Without it, coordinators would need to manually create each transit leg, increasing the risk of delays and errors in time-critical deliveries. The FS-IL contains the same material as the FS-FL; other details (pickup/dropoff locations, contacts) are determined by the flow version. A new flag sap/cgt/ext/shipment/intermediate_shipment/create (2511) also enables creating subsequent intermediate product shipments programmatically.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM005	Active/Inactive state management	All shipment types support the Active/Inactive state model introduced in release 2409 for follow-on order handling. When a follow-on treatment order is created, parent order's shipments are marked Inactive along with all other entities. No changes are allowed on Inactive shipments, and all inbound events are automatically routed to the active order's shipments. The State flag is displayed in the header of all Manage Shipments actions.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM006	Shipment Information APIs	The Shipment Information V2 API (GET /Shipments) retrieves all shipments with COI ID, Order ID, and Treatment ID. The Shipment Collection Information API (GET /Shipments/Collection) retrieves shipment collection information.

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					/ShipmentCollectionView) retrieves collection details. Shipment Approval Info API (POST /UpdateShipmentApprovalInfo) updates shipment approval information. Release 2502 significantly enhanced the Shipment Information API with new \$expand options (contacts, business partners, documents, courier references, and custom texts) and 40+ new fields. Release 2506 expanded support to BKS, IS, and FS-IL shipment types. These updates enable treatment center platforms and logistics dashboards to display real-time shipment status across the entire therapy journey, giving care coordinators a unified view of where every patient's material is at any moment. Shipment API also returns all milestones, including custom milestones.
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM007	Manage Shipments apps	A family of 5 dedicated Fiori apps provides the primary user interface for managing shipments across all five shipment types – paralleling the Manage Orders app (TOM002) for treatment orders and the Manage Processing Activities app (PAM001) for manufacturing. Each shipment type has its own set of apps: Manage Shipments – Biospecimen, Manage Shipments – Final Product (covering both FS-IL and FS-FL), Manage Shipments – Intermediate Product, and Manage Shipments – Biospecimen Kit. All Manage Shipments apps share a consistent UI structure: a tabbed layout (converted from scroll-based in release 2409 for performance) with tabs for general information, patient details, location, shipment milestones (standard & custom), logistic information (courier booking, shipper tracking), materials and subunits, business partners, custom texts, documents, and logs (change log, document log, e-signature log). Field visibility is configurable per shipment type and order type via field control profiles. The apps provide navigation to related entities – open tasks, exceptions, approval details, and the associated ERP document (purchase order, sales order, or stock transfer order).
Supply Chain	Shipment & Logistics	Shipment Lifecycle Management	SLM008	Subsequent shipment creation for Intermediate Shipments	Users can now request and create subsequent Intermediate Product Shipments, both via event trigger and manually through the UI, after Flow Version (FV) confirmation.
Supply Chain	Shipment & Logistics	Shipment Processing	SPR001	Biospecimen shipment processing	A suite of apps manages the biospecimen receipt workflow: Manage Incoming Biospecimen (view/manage collection details), Confirm Biospecimen Receipt (conduct condition and COI confirmation checks, subunit receipt confirmation), and Manage Biospecimen Disposition (track disposition status for subunits). The biospecimen receipt step is the critical handoff point where patient-collected material (e.g., leukapheresis product) arrives at the manufacturing site – shipping condition checks verify temperature requirements were maintained during transport, and COI confirmation ensures the material

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					belongs to the correct patient before manufacturing begins. Collection details include the Vendor Batch (2506) for batch traceability. Biospecimen shipment events support material batch-level custom texts (and Planned Dropoff Date (2502). The biospecimen receipt event from SAP S/4HANA updates batch number and actual delivery date in SAP Advanced Therapy Orchestration.
Supply Chain	Shipment & Logistics	Shipment Processing	SPR002	Intermediate product shipment processing	Four dedicated apps introduced in release 2404 manage the intermediate product workflow: Allocate Batch Intermediate Product Shipments, Confirm Intermediate Product Shipment Pickup, Confirm Intermediate Product Shipment Receipt (shipping conditions and COI confirmation), and Manage Intermediate Product Shipment Disposition. Each app supports document change logs, e-signature logs, and reason codes. Each authorization roles (App + Write pairs for receipt, disposition, allocate, pickup) control access to these Intermediate product shipments support multi-site manufacturing where, for example, viral vector production occurs at one CMO and cell transduction at another. Each intermediate shipment ensures the partially processed material maintains chain of identity and proper temperature control during the inter-site transfer.
Supply Chain	Shipment & Logistics	Shipment Processing	SPR003	Finished product shipment processing	The finished product workflow includes Allocate Batch to Finished Product Shipments (allocate/deallocate batches with subunit details), Confirm Finished Product Pickup (verify shipper serial numbers and product packing), and the infusion location management feature (2409) for viewing and updating the Product Administration Center on FS-FL shipments – the final destination where the patient receives the therapy, may differ from the ordering treatment center. For products with parallel flows, the flow step identifies previous processing activity for allocation and deallocation (2502). Approval details include a “Decision By” field showing who made the batch release decision (2409) providing transparency into the quality authority at the point of final delivery.
Supply Chain	Shipment & Logistics	Shipment Processing	SPR004	Transit leg shipment processing	Finished Product Transit Leg shipments support the logistics lifecycle: manual shipment and receipt confirmation on the Shipment and Receipt tab, batch allocation/deallocation (from PA yield batch or pre shipment batch with “Shipped” status validation), PGI/PGR and reversals, material batch disposition quality inspection, and batch release decision history in SAP Batch Release Hub. Transit legs model the intermediate hops in CGT distribution – for example, cryopreserved CAR-T product may ship from the manufacturing CMO to a regional distribution hub for the final leg to the treatment center, with full goods movement and quality inspection at each intermediate

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					point. The ERP tab displays STO data including sending/receiving plants and create/update/cancel response details. Update via inbound event sap/cgt/ext/shipment/multi_type/update (2502).
Supply Chain	Shipment & Logistics	Shipment Processing	SPR005	Shipment cancellation workflow	A universal cancellation workflow (introduced in 25 applies to all 5 shipment types across 7 logistics and Request Cancellation -> Trigger Workflow -> Approve/Reject -> Cancel ERP Document -> Send to Treatment Center. Cancellation can be initiated from UI or via the generic inbound event sap/cgt/shipment/cancellationRequest/update. The outbound response event sap/cgt/tmp/shipment/cancelStatus confirms the Cancellation is only allowed from "Shipment Request" "Shipment Scheduled" status – this restriction prevents cancellation of shipments already in transit carrying patient-specific material, where cancellation would be a logistical crisis and potential material loss. Six new authorization roles were introduced for BKS, FS-IL, and cancellation (request + approve/reject each).
Supply Chain	Shipment & Logistics	Shipment Processing	SPR006	Receive Milestone updates	Introduced support for receiving and updating both standard and custom milestones in CGTO from external systems via the new sap/cgt/ext/shipment/milestone/update event.
Supply Chain	Shipment & Logistics	Shipment Processing	SPR007	Trigger Sales Order update	CGTO now consistently triggers S/4 Sales Order Update events for any updates made to FS-FL shipments, including to such as Courier Actual Delivery Date, Product Acceptance Status, and FP Shipment Texts. This enhancement ensures reliable synchronization between CGTO and S/4, keeping ERP documents up-to-date as shipment changes in real time.
Supply Chain	Shipment & Logistics	Shipment Processing	SPR008	Manual Biospecimen Collection Management	Ability to manually create, edit, assign, and unassign Biospecimen Shipment Collections from the CGTO enabling standalone-mode operation and correcting workflows independent of inbound events.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT001	Courier booking	The Book Courier button on shipment apps sends a booking request via the Shipment Courier Booking outbound event to the courier system. Prerequisites include confirmed schedule (optional), specified sender/receiver organizations with contacts, and shipment materials. Gross weight is calculated based on a configurable quantity type (triggered at flow version confirmation for requested materials, at material allocation for allocated materials). The courier system returns shipper serial numbers, temperature monitoring serial numbers, and tracking links. CGT shipments use specialized courier services (e.g., World Courier, FedEx Custom Critical) because each shipment carries irreplaceable patient-specific material under strict temperature control, making courier booking a critical

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					where errors can result in therapy loss. Biospecimen Shipment courier booking was added in release 2506.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT002	Courier booking fields (2506)	Release 2506 enriched the courier booking and shipment screens with new fields: Pickup Start/End Time and Dropoff Start/End Time (preferred courier time slot sourced from organization location master data), Organization Names (alternate names for shipper and consignee organizations), Temperature Condition Code (required storage temperature for materials), and Healthcare Organization PO Number (hospital purchase order identifier). All fields are included in the courier booking event payload sent to courier systems. The Temperature Condition Code is particularly important for CGT logistics because it communicates the exact storage requirement (cryogenic, 2-8C, or ambient) to the courier, ensuring the correct shipper type and temperature monitoring equipment are dispatched for each shipment.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT003	Failed courier booking handling	The Courier Booking Response event was enhanced in release 2506 to receive failure messages from courier systems. A new Failed Courier Bookings section displays the Logistics tab of Manage Shipments apps (BS, FLS, BKS, inbound/outbound logistics, and confirmations) showing Request ID, Error Code, Error Description, Error Type, Error Log ID, and Created/Changed By/On. This provides visibility into booking failures that previously required investigation in external courier systems - particularly important in CGT where a failed booking for biospecimen pickup can delay the entire therapy timeline as cells lose viability with each hour of delay.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT004	Multiple-shipper support	Finished product shipments support multiple shipper shipments, each with its own serial number, temperature monitoring device number, and tracking link. This supports CGT scenarios where a single patient's therapy involves multiple containers requiring separate cryoshipper independent temperature monitoring. Users can select requested shipper count to the courier partner and shipper serial numbers to collections/subunits. The "Temperature Monitoring Link" was renamed to "Serial Tracking Link" and moved from shipment level to subunit level. Controlled by feature flag (name unspecified). Impacts: Manage Shipment FP, Manage Outbound Logistics, Allocate Batches, Confirm FP Pickup, and Monitor. Introduced in release 2404.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT005	Transportation lane determination	The Determine Transportation Lane button on shipment apps automatically selects the logistics route based on configured transportation lane rules. Rules match material type + material group (or transportation packaging type) to select the appropriate lane. When multiple rules match, the most specific rule wins. The determined lane populates the Logistics tab with collection and route information. Automatic lane determination

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					ensures that each shipment is routed through the correct carrier for its material type – cryopreserved products require validated cryogenic shipping lanes, while fresh products need same-day courier routes with ambient temperature control. Transportation lanes and rules are maintained as master data in the Manage Transportation Lanes and Manage Transportation Lane Rules applications.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT006	Courier milestone tracking	Courier booking milestones from SAP S/4HANA Transportation Management are sent to SAP Advanced Therapy Orchestration via the Courier Booking Milestone Update event. Currently the VLPOD (proof of delivery) milestone is built into the standard delivery, with the framework extensible via the SAP S/4HANA add-on to support additional milestone types. The Courier Booking Response event also carries real-time courier data (waybill, ASN Number, Carrier ID) from Freight Order/Booking. These events are triggered only when the Freight Order/Booking has a linked COI ID. Real-time milestone tracking is essential for CGT logistics coordinators who need to know exactly when a shipment was picked up, in transit, or delivered – delays measured in hours can render time-sensitive cell therapy products unusable. Introduced in release 2502.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT007	Courier instructions in shipment logistics and courier booking	Integrate courier instructions into shipment workflow. Instructions configured by order type are auto-populated into shipments during creation, displayed in the Advanced Courier Instructions table within the Logistics tab for all five shipment types. Users can edit instructions and selectively include/exclude instructions during booking via checkboxes. Selected instructions are transmitted to courier partner systems via Book Courier API. Instructions persist through order cloning and subsequent shipment creation, with full change log tracking and API support via Shipment Information System.
Supply Chain	Shipment & Logistics	Courier & Transportation	CRT008	Courier booking response	Process courier booking confirmation responses from external courier partner systems. The Courier Booking Response event receives booking status, tracking information, and courier milestone updates after a request is processed. Enhanced in release 2605 to support optional courier instructions from courier systems to include instruction IDs and values that are automatically added or update the shipment's Additional Courier Instructions table in the Logistics tab. Courier-originated instructions do not require pre-configuration in Configure Courier Instructions, enabling courier partners to send special handling instructions such as confirmed pickup with routing codes, or handling alerts. CGTO validates instruction ID length and either updates existing instruction values or adds new instructions to the shipment. Users can edit received instruction values and selectively include or exclude them during subsequent courier booking operations. Instructions are accessible via the Advanced Courier Instructions table.

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					across all shipment types via Manage Shipments a Manage Inbound/Outbound Logistics apps, and the Shipment Information API
Supply Chain	Shipment & Logistics	Business Partner Management	BPM001	Business Partners tab	A dedicated Business Partners tab on shipment ap enables viewing, adding, and deleting business par per shipment. Partners are categorized by functio Sold-To, Goods Recipient, Ship-To, Vendor, Supplier specified by organization, location type, business p role, business partner ID, and business partner fun. The tab was introduced for FS-FL in release 2409, t extended to BS, BKS, IS, Inbound Logistics, and Ou Logistics in release 2502. Partners received via inb events are validated against available master data. business partner assignment is critical for ERP doc creation – the Vendor on a biospecimen shipment determines the purchase order partner, and the Bil To/Sold-To on a finished product shipment drives t order in the therapy network.
Supply Chain	Shipment & Logistics	Business Partner Management	BPM002	Business partner determination on schedule confirmation	During schedule confirmation, the system checks f minimum required business partners per shipment Vendor/Supplier for biospecimen shipments, Bill To To/Ship To for finished product transit leg shipmer required partners are missing, business rules (BRF attempt automatic determination. A single unique populates the Business Partners tab; multiple mat result in unsuccessful determination. The schedule confirmed regardless of determination success, en that time-critical therapy scheduling is never block incomplete business partner data – missing partn be resolved after confirmation without delaying the patient's treatment timeline. First introduced for FS release 2409, extended to BS and FS-IL in release 2
Supply Chain	Shipment & Logistics	Business Partner Management	BPM003	Automatic business partner function determination	Across all five shipment types, the system automa determines the business partner function from configuration when it is not explicitly provided. Exp provided functions are accepted as-is for backwar compatibility. Invalid functions trigger warnings bu still accepted. When not provided, the system dete the function from configuration, with fallback to BF when configuration is missing. A new Business Par Function field with F4 help (showing options by Organization, Location Type, or Role) was added to Shipments apps. Introduced in release 2511.
Supply Chain	Shipment & Logistics	Business Partner Management	BPM004	Business partner synchronization to SAP S/4HANA	When business partners are updated for any shipr type, the updated list is passed to SAP S/4HANA v corresponding ERP document event (Purchase Or Update, Sales Order Update, Stock Transport Orde Update). SAP S/4HANA reads the updated partner applies delta changes. This bidirectional synchroni ensures that the ERP documents reflect the curre business partner assignments in SAP Advanced Th

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					Orchestration – essential because CGT therapy net involve frequent partner changes when treatment c redirect deliveries or CMOs change mid-process. B partner arrays were added to inbound BS Create/L events in release 2502.
Supply Chain	Shipment & Logistics	Inbound & Outbound Logistics	IOL001	Manage Inbound Logistics app	A consolidated view for managing inbound logistics: biospecimen and intermediate product shipments. app provides general information, patient info, loca shipment dates, logistics info, shipment/receipt confirmation, approval details, PPF events, docum and change/document/e-signature logs. Users car determine transportation lanes, book couriers, and manage the full inbound logistics workflow from a : screen. This consolidated view is critical for CGT lo coordinators who manage the arrival of patient-spe materials at manufacturing sites – they need to simultaneously track courier status, verify shipping conditions, confirm chain of identity, and trigger gc receipt without switching between multiple apps. Enhanced in release 2404 with intermediate produ shipment support and in 2502 with cancellation wc and business partner management.
Supply Chain	Shipment & Logistics	Inbound & Outbound Logistics	IOL002	Manage Outbound Logistics app	A consolidated view for managing outbound logisti across finished product (final leg and transit leg) an intermediate product shipments. The app includes same capabilities as inbound logistics plus the infu location feature for FS-FL shipments (view and upc Product Administration Center, introduced in 2409 outbound logistics view is where the final critical h occurs – the manufactured therapy product leaves controlled environment of the manufacturing site a enters the courier network, making this the last po where logistics coordinators can verify that batch allocation, courier booking, and temperature requi are all aligned before the product ships to the patie Enhanced progressively with intermediate product (2404), transit leg support (2409), cancellation wc (2502), and business partner management (2502)
Supply Chain	Shipment & Logistics	Inbound & Outbound Logistics	IOL003	Multiple contact persons	Alternate contact persons (name, email, phone) ca managed for pickup, delivery, and courier locations all shipment management apps and both inbound : outbound logistics apps. Users can add, update, or alternate contacts. Contact persons must be maini the Manage Contact Persons master data app bef can be added to a shipment. This supports CGT log scenarios where multiple contacts need to be notif collection centers, manufacturing sites, and treatr facilities. Introduced in release 2409.
Supply Chain	Shipment & Logistics	Inbound & Outbound Logistics	IOL004	COI confirmation on receipt	When a shipment is received, the COI Confirmation transitions from Pending (ST042) to Confirmed (S Rejected (ST043) based on automated COI checks

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					shipping conditions check or the COI confirmation fails, an automatic exception is generated with source "Internally Generated", Priority "High", and Type "Error". This applies to biospecimen shipments, intermediate product shipments, and finished product transit leg shipments. The COI confirmation mechanism is a compliance control ensuring that the correct patient material is being processed at each stage – in auto CGT, a chain of identity failure means the wrong patient cells could be manufactured or administered, with potentially fatal consequences.
Supply Chain	Scheduling	Slot Management	SMG001	Manage Manufacturing Slots app	A dedicated app introduced in release 2511 providing comprehensive slot management platform for manufacturing capacity planning. The app enables Managers to define manufacturing time slots per patient (organization, location type, plant name), select which materials can be processed in those slots, and manage capacity over time. Each slot definition must be unique plant-material combination. Manufacturing capacity of the biggest bottlenecks in CGT – cleanroom suite specialized equipment, and trained operators are scarce and each patient's therapy requires dedicated capacity that cannot be shared or batched with other patients making slot management essential for balancing patient demand against limited resources. Materials within definition can be activated and deactivated independently. Access is controlled by CGT_SlotMaintenance_Read (read) and CGT_SlotMaintenance_Write_Srv (write).
Supply Chain	Scheduling	Slot Management	SMG002	Calendar view	The Manage Manufacturing Slots app displays a calendar view showing total, available, hard-booked, and soft-booked slots per day per material. Users can filter by material, period, and date to focus on specific capacity needs. The visual representation makes it easy to identify days with full capacity, available capacity, or overcommitment at a glance. This replaces what was previously a data-only scheduling interface with a visual capacity planning tool that mirrors the way CGT Plant Managers actually think about their manufacturing as a calendar of patient slots that must be carefully sequenced to avoid cleanroom conflicts, staff overcommitment, or gaps in equipment utilization.
Supply Chain	Scheduling	Slot Management	SMG003	Capacity management	Plant Managers can increase or decrease slot capacity using the Update Capacity function. The system enforces a minimum capacity rule: capacity cannot be reduced below the number of already-booked (hard or soft) slots, preventing overbooking situations. Capacity changes are reflected in real-time across the calendar view and availability calculations, ensuring that scheduling proposals generated by the Get Proposals API always reflect the current capacity state. In CGT, where each manufacturing slot represents a commitment to a patient whose cells have already been collected and

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					degrading over time, overbooking is not just an operational inconvenience but a potential therapy failure.
Supply Chain	Scheduling	Slot Management	SMG004	Order allocation and deallocation	Treatment orders can be manually allocated to or deallocated from hard-booked manufacturing slots. Allocation assigns a specific treatment order to a slot, while deallocation removes the assignment. Changes from allocation and deallocation are reflected in real-time in the calendar view. This manual control complements the automated scheduling proposal booking workflow, enabling Plant Managers to handle exceptions or priority changes directly – such as booking a critically ill patient's order ahead of a stable patient's scheduled slot, or reallocating a slot when a biospecimen collection is delayed.
Supply Chain	Scheduling	Slot Management	SMG005	Slot change logging	All slot management actions are tracked in change logs with user ID, timestamp, and old/new values. Logged actions include: slot definition creation, material activation/deactivation, capacity updates (increase/decrease), and order allocation/deallocation. This audit trail ensures full traceability of capacity management decisions, supporting GxP-compliant manufacturing scheduling.
Supply Chain	Scheduling	Schedule Confirmation	SCF001	Manage Schedules app	The central app for reviewing, editing, and confirming treatment order schedules. All milestones are displayed in a single table, differentiated by shipment type and by flow step, giving Treatment Order Managers a clear end-to-end view of the therapy timeline from biospecimen collection through manufacturing to final delivery. Users can view and edit milestones dates for each entity, which trigger ERP document creation, which validates the schedule and triggers ERP document creation in SAP S/4HANA (Purchase Order for BS, Sales Order for FS-FL and Stock Transport Order for IS and FS-IL, Process/Production Order for PA). Trigger ERP document creation (earlier called Schedule confirmation) is the final moment where the therapy plan becomes operational reality, committing resources across the entire supply chain. Individual trigger ERP document creation for specific entities (BS, FP, or PA) as standalone SAP S/4HANA requests is supported (2404).
Supply Chain	Scheduling	Schedule Confirmation	SCF002	Scheduling algorithm	The scheduling algorithm generates proposals based on slot availability and configured lead times. The user provides a preferred biospecimen pickup date (or the system defaults to today). The system calculates the manufacturing milestone date using lead times of preceding milestones, checks slot availability at the derived date, and if available, calculates all subsequent milestone dates to generate a schedule proposal. If unavailable, the system finds the next available slot, recalculates all dates, and generates alternative proposals. Both single-plant and multi-plant scheduling flows

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					supported. The algorithm for scheduling requests for treatment orders can also support for a limited to a specified range of milestones (partial scheduling), in addition to complete scheduling.
Supply Chain	Scheduling	Schedule Confirmation	SCF003	Automatic schedule confirmation	Schedules can be auto-confirmed in two scenarios without manual user action: (1) for new orders, when the flow version is confirmed (requires PPF event configuration validated planned dates), and (2) for existing order confirmed flow version, when subsequent shipment processing activities are created via events with the autoConfirmSchedule flag set to true. This flag has been added to the Biospecimen Shipment Creation, Finished Product Shipment Creation (Final Leg), and Processing Activity Create events. This capability reduces manual steps in high-volume orchestration scenarios where manufacturers process dozens of concurrent patient therapies. Introduced in release 2502.
Supply Chain	Scheduling	Schedule Confirmation	SCF004	Schedule approval and hard booking	The Approve Schedule button in the Confirm Schedule app (controlled by feature flag ft_support_scheduling) triggers hard booking of manufacturing slots when proposed schedule dates are accepted. Hard-booked slots are permanently reserved until cancelled or rescheduled, replacing the temporary soft-booked reservations. The two-stage model mirrors CGT scheduling practices where treatment centers initially explore dates (soft book) before committing to a collection date with the patient (hard booking) – once hard-booked, the manufacturing slot is guaranteed and the patient can be prepared for leukapheresis. If dates are changed after soft booking, the system re-checks slot availability and generates a new proposal. The approval triggers the outbound event <code>sap/cgt/order/treatment_order/related_entities_w/confirm</code> . The Scheduling section containing milestones, scheduling request ID, proposal ID, and success/error message is displayed. Introduced in release 2511.
Supply Chain	Scheduling	Schedule Confirmation	SCF005	Business partner determination on confirmation	During schedule confirmation, the system checks for a minimum required business partners per shipment and attempts automatic determination via business partner determination (BRFplus) when partners are missing. For biospecimen shipments, the system checks for Vendor/Supplier partners. For finished product transit leg shipment checks for Bill To, Sold To, and Ship To partners. A single unique match auto-populates the Business Partner field. Multiple matches result in unsuccessful determination. The schedule is confirmed regardless of determination outcome, ensuring the process is not blocked. Enhanced progressively from FS-FL (2409) to BS and FS-IL (2502).
Supply Chain	Scheduling	Schedule Confirmation	SCF008	Scheduling Request- Cancellation	This action allows users to cancel scheduling requests. Supports complete and partial cancellation. Indirect Schedule Request cancellation is also supported via Treatment Order or Processing Activity cancellation.

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Supply Chain	Scheduling	Scheduling APIs	SCA001	Get Proposals API	The POST /getProposals API generates treatment milestone proposals by evaluating slot availability, milestone configurations, and lead time parameter. Mandatory inputs include Therapy and Country (for version determination) and Location (Collection Center or Biospecimen Shipment Pickup, or Finished Product Shipment Delivery). Optional inputs include Input Milestone Date Time (defaults to current date), Input Milestone Type (Biospecimen Shipment Pickup or Biospecimen Collection), and Flow Version (if not provided up to 2 flow versions are determined via business decisions). The API returns a maximum of 30 proposals per request, including standard & custom milestones enabling treatment center platforms and patient scheduling systems to programmatically find the earliest available therapy slot for a patient without manual intervention in the CGTO UI – essential for scaling operations from dozens to hundreds of concurrent patients. Introduced in release 2511. Create scheduling requests for treatment orders can also support a list of a specified range of milestones (partial scheduling) in addition to complete scheduling.
Supply Chain	Scheduling	Scheduling APIs	SCA002	Book Processing Slots API	The POST /bookProcessingSlots API reserves one manufacturing slots for a scheduling request. Mandatory inputs are schedulingRequestId and proposalId (from prior Get Proposals call) and bookingType (currently limited to soft). Optional inputs include Milestones (override proposal dates) and COI ID. Reservations created via this API are soft-booked and expire automatically if not confirmed within the defined time frame, aligning with a two-stage booking model where soft booking converts to hard booking only upon schedule approval in the CGT Schedules app. The automatic expiration of unconfirmed soft bookings prevents treatment centers from indefinitely holding scarce manufacturing capacity that other patients need – a critical safeguard as CGT manufacturers scale operations. Introduced in release 2511. Enhanced the Book Slots API to support rescheduling by allowing an optional Reference Scheduling Request ID.
Supply Chain	Scheduling	Scheduling APIs	SCA003	Cancel Scheduling Request	This event allows users to cancel scheduling requests for treatment orders. Supports complete and partial cancellation: Supported also via Confirm Schedule
Supply Chain	Scheduling	Slot Calendar	SCC001	View Available Slots	Users can view available slots directly from the app through “View Slots” app. Slot booking is also possible through the application
Master Data & Configuration	Business Configuration	Milestone Configuration	EPC007	Configure Milestones app	The Configure Milestones application enables users to create and manage custom milestones that reflect events within operational workflows. Custom milestones can be associated with core entities such as Treatment Orders, Processing Activities, or Shipments. When a milestone is mapped to a Shipment, it can be further

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					configured based on the shipment type (for example Biospecimen shipments). Each shipment related record also supports timezone configuration, enabling users to choose whether milestone timestamps are based on Pickup Timezone, or the Delivery Timezone. By default, milestones are configured to use the Pickup Timezone, ensuring consistent timestamping unless otherwise specified. The application also has a view of system milestones, which is only for display and cannot be created or edited.
Cross Applications	Treatment Management	Order Scheduling & Confirmation	OSC005	Schedule Requests	Users can now view all scheduling requests associated with a Treatment Order in a dedicated table. New "Scheduling Requests" tab/section displaying request details (ID, status, expiry date, slots booked, created/modified info, etc.), with default sorting by "Created On" field. Ability to drill down into each scheduling request for read-only detailed information including milestones and key reference attributes.
Supply Chain	Scheduling	Schedule Confirmation	SCF006	Scheduling Request – Rescheduling	Users can now efficiently reschedule treatment orders ensuring slot capacity is correctly managed and freed without manual intervention. To achieve this Rescheduling, users need to get Slots (either via Get Slots API or View Slots app). They can select any proposal & use enhanced Book Slots API to book it, additionally passing an optional Reference Scheduling Request ID (i.e. Scheduling Request ID). When rescheduling with a reference, the system can automatically obsolete (or partially obsolete) the prior scheduling request and associated future slots. Error messaging improved to notify users if the reference scheduling request is already cancelled/obsolete. No change to user interface layout, but users will see status updates for obsolete/partially obsolete scheduling requests and relevant error/warning popups during the rescheduling process. Impact: . Applicable to both complete and rescheduling scenarios. No impact if the reference scheduling request is not provided (manual cancel still possible).
Application Platform & Infrastructure	Integration	Scheduling APIs	SAH001	Approval & Hard Booking	Added ability to approve treatment schedules directly through the "Manage Schedule" app via an "Approve Schedule" button, when integrated with an external scheduling system. Configuration available for administrators to map external scheduling API endpoints (via new "Scheduling API Configuration" rule provider in destination setup). Upon approval, system automatically triggers a configurable external API call to create a detailed order and shipment payload. Enhanced error handling: clear technical error messaging if the destination is missing or API call fails; user receives immediate feedback from the external system response. "Approve Schedule" button in "Scheduling Requests" section disabled when external scheduling integration is active. No impact for users not utilizing external scheduling.

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					integration; backward compatibility unaffected. Im Streamlines external schedule approvals and impr user workflow for sponsored therapy orders requir third-party scheduling coordination.
Supply Chain	Scheduling	Schedule Confirmation	SCF007	Schedule Requests	Users can now view all scheduling requests associ with a Treatment Order in a dedicated table. New “Scheduling Requests” tab/section displaying req details (ID, status, expiry date, slots booked, created/modified info, etc.), with default sorting by “Created On” field. Ability to drill down into each scheduling request for read-only detailed informat including milestones and key reference attributes. Scheduling Request supports the Cancel, Approve Reschedule actions
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI015	Subcontracting Purchase Order Lifecycle	The SAP Advanced Therapy Orchestration Subcon Purchase Order (SubCon PO) Integration enables automated end-to-end management of CGT manufacturing workflows . SubCon POs are autom created in S/4HANA upon IS or FS-IL shipment confirmation in CGTO, with ERP Data tab reflecting time PO status. The integration supports PO updat (dates, partners, quantities, status), cancellations, lifecycle management of Goods Receipts via IM or S/4 HANA, including partial PGR, PGR reversal, and validation. Component consumption and reversal e update Processing Activity milestones, Chain-of-Custody/Chain-of-Identity records, and batch consumption status. Quality inspection and Usage Decisions – posted via QA32, QA11, or SAP Batch R Hub – trigger disposition events to CGTO and down Treatment Ordering Platforms via Event Mesh, corr the patient-centric traceability chain.
Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI016	WBS Integration with CGTO	The SAP Advanced Therapy Orchestration WBS Int links S/4HANA Work Breakdown Structure elemen CGTO therapy supply chain entities for real-time pa therapy monitoring, chain-of-identity visibility, and traceability. WBS elements and Network numbers (manually entered as free-text fields across CGTO e Treatment Order. WBS values automatically propag from S/4HANA Sales Order responses to FS-FL shi then upstream to the Treatment Order, and visible t linked entities. For S/4HANA document creation, w WBS is included in CGTO event payloads, Purchase are created by converting matching Purchase Req filtered by WBS, delivery date, vendor, plant, and m Process/Production Orders are created by convert Planned Orders using WBS, dates, plant, and mater Sales Orders are assigned the WBS with the value e back to CGTO for propagation. WBS mismatch betw CGTO and S/4HANA raises validation errors, and c to an already-set Treatment Order WBS require exp user confirmation to prevent downstream disruptio

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Application Platform & Infrastructure	Integration	SAP S/4HANA ERP Document Integration	EDI017	Decentralized EWM intergation with CGTO	When goods movements are executed in decentral EWM, material document creation requests are sent to S/4HANA, which triggers the corresponding outbound events to CGTO with mapped Handling Units (HUs) and Sub-units (SUs). The default HU-to-SU sequential logic applies across all supported document types: Purchase Orders, PGR completion and inspection lot release trigger PGR and disposition events to CGTO; Process Orders, PGI completion, PGR completion, and inspection lot release each trigger the corresponding PGR, and disposition events to CGTO; for Sales Order batch picking in outbound delivery triggers an Allocation event, followed by a PGI event on goods issue completion for Stock Transport Orders, PGI completion, PGR completion, and inspection lot release trigger the respective PGI, PGR, and disposition events to CGTO. In all cases, HU-to-SU mapping is included in the event payload ensuring full traceability of serialized materials across the decentralized warehouse management landscape.